

模式识别与计算机视觉

Warm Up

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Perspective distortion

Finite projective camera

$$\mathbf{P} = \begin{bmatrix} \alpha_x & s & p_x \\ 0 & \alpha_y & p_y \\ 0 & 0 & 1 \end{bmatrix} \left[\mathbf{R} \mid -\mathbf{RC} \right]$$



What does this matrix look like if the camera and world have the same coordinate system?

Finite projective camera

The pinhole camera and all of the more general cameras we have seen so far have “*perspective distortion*”.

$$\mathbf{P} = \begin{bmatrix} \alpha_x & s & p_x \\ 0 & \alpha_y & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$


Perspective projection from
(homogeneous) 3D to 2D coordinates

Perspective distortion



long focal length



mid focal length



short focal length

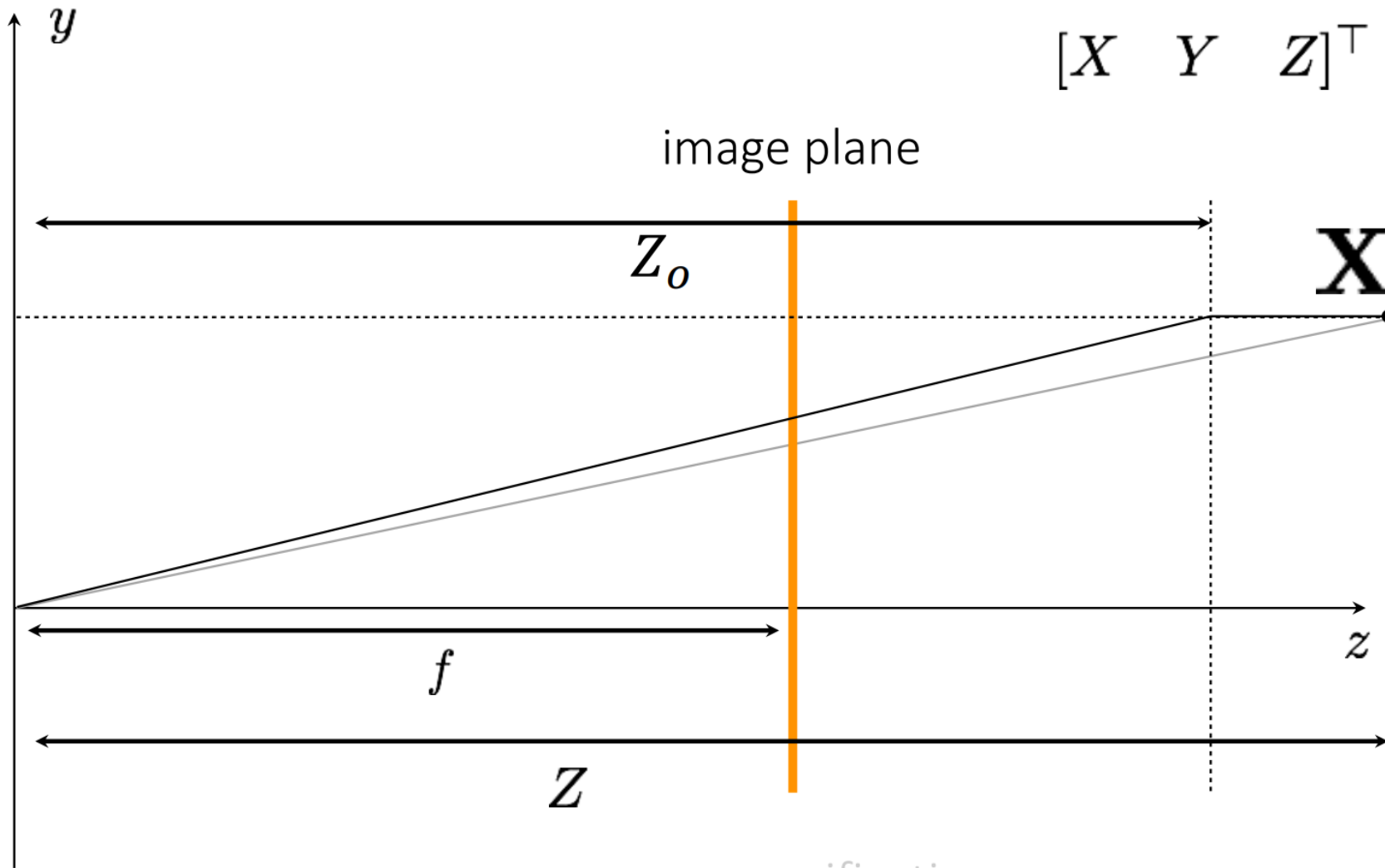
Vertigo effect

Named after Alfred Hitchcock's movie

- also known as “dolly zoom”



Weak perspective vs perspective camera



$$[X \ Y \ Z]^T \mapsto [fX/Z_o \ fY/Z_o]^T$$

- magnification does not change with depth
- *constant* magnification depending on f and Z_o

magnification
changes with depth

$$[X \ Y \ Z]^T \mapsto [fX/Z \ fY/Z]^T$$

Comparing camera matrices

Let's assume that the world and camera coordinate systems are the same.

- The *perspective* camera matrix can be written as:

$$\mathbf{P} = \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

- The *weak perspective* camera matrix can be written as:

$$\mathbf{P} = \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & Z_o \end{bmatrix}$$

When can we assume a weak perspective camera?

1. When the scene (or parts of it) is very far away.

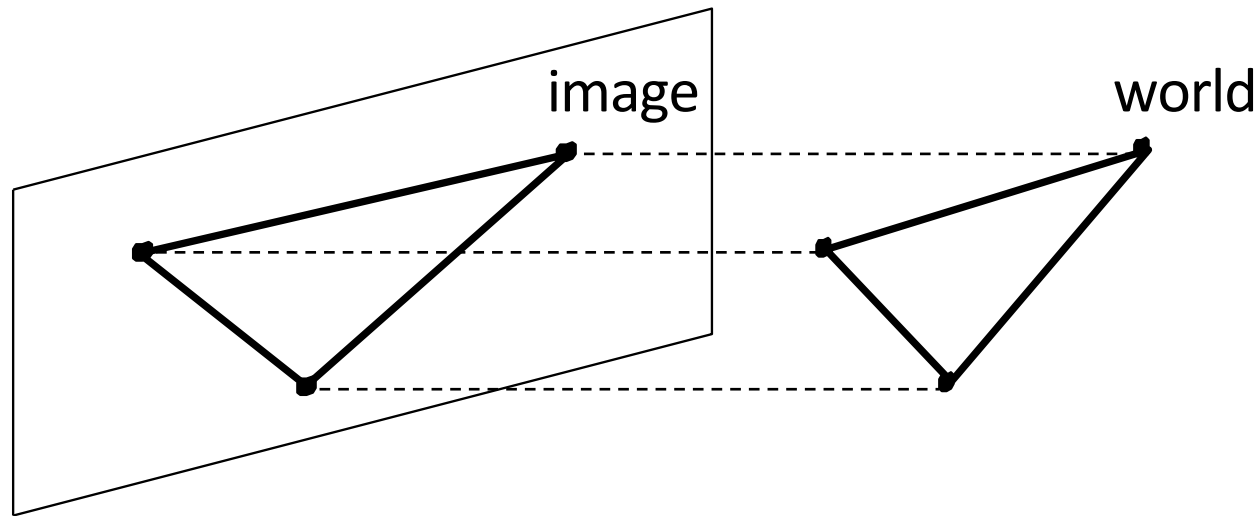


Weak perspective projection applies to the mountains.

Orthographic camera

Special case of weak perspective camera where:

- constant magnification is equal to 1.
- there is no shift between camera and image origins.
- the world and camera coordinate systems are the same.

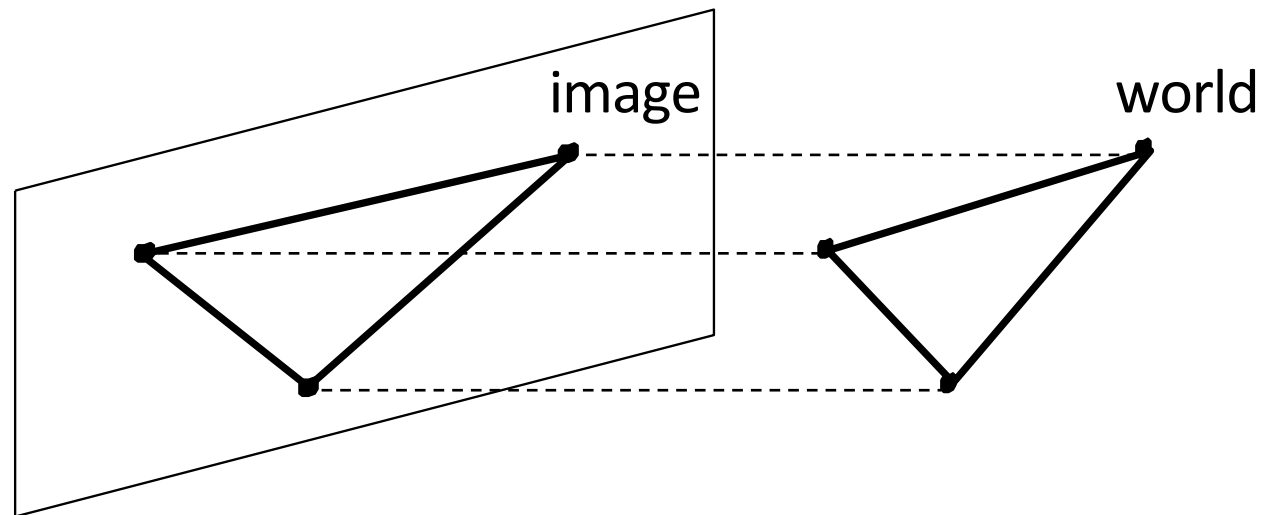


What is the camera matrix in this case?

Orthographic camera

Special case of weak perspective camera where:

- constant magnification is equal to 1.
- there is no shift between camera and image origins.
- the world and camera coordinate systems are the same.



$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Geometric camera calibration

	Structure (scene geometry)	Motion (camera geometry)	Measurements
Camera Calibration (a.k.a. Pose Estimation)	known	estimate	3D to 2D correspondences
Triangulation	estimate	known	2D to 2D coorespondences
Reconstruction	estimate	estimate	2D to 2D coorespondences

Geometric camera calibration

Given a set of matched points

$$\{\mathbf{X}_i, \mathbf{x}_i\}$$

point in 3D
space

point in the
image

and camera model

$$\mathbf{x} = \mathbf{f}(\mathbf{X}; \mathbf{p}) = \mathbf{P}\mathbf{X}$$

projection
model

parameters

Camera
matrix

Find the (pose) estimate of

P

We'll use a **perspective** camera
model for pose estimation

Mapping between 3D point and image points

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

What are the unknowns?

Mapping between 3D point and image points

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \text{---} & \mathbf{p}_1^\top & \text{---} \\ \text{---} & \mathbf{p}_2^\top & \text{---} \\ \text{---} & \mathbf{p}_3^\top & \text{---} \end{bmatrix} \begin{bmatrix} | \\ \mathbf{X} \\ | \end{bmatrix}$$

Heterogeneous coordinates

$$x' = \frac{\mathbf{p}_1^\top \mathbf{X}}{\mathbf{p}_3^\top \mathbf{X}} \quad y' = \frac{\mathbf{p}_2^\top \mathbf{X}}{\mathbf{p}_3^\top \mathbf{X}}$$

(non-linear relation between coordinates)

How can we make these relations linear?

How can we make these relations linear?

$$x' = \frac{\mathbf{p}_1^\top \mathbf{X}}{\mathbf{p}_3^\top \mathbf{X}} \quad y' = \frac{\mathbf{p}_2^\top \mathbf{X}}{\mathbf{p}_3^\top \mathbf{X}}$$

Make them linear with algebraic manipulation...

$$\mathbf{p}_2^\top \mathbf{X} - \mathbf{p}_3^\top \mathbf{X} y' = 0$$

$$\mathbf{p}_1^\top \mathbf{X} - \mathbf{p}_3^\top \mathbf{X} x' = 0$$

Now we can setup a system of linear equations
with multiple point correspondences

$$\mathbf{p}_2^\top \mathbf{X} - \mathbf{p}_3^\top \mathbf{X} y' = 0$$

$$\mathbf{p}_1^\top \mathbf{X} - \mathbf{p}_3^\top \mathbf{X} x' = 0$$

How do we proceed?

$$\mathbf{p}_2^\top \mathbf{X} - \mathbf{p}_3^\top \mathbf{X} y' = 0$$

$$\mathbf{p}_1^\top \mathbf{X} - \mathbf{p}_3^\top \mathbf{X} x' = 0$$

In matrix form ...

$$\begin{bmatrix} \mathbf{X}^\top & \mathbf{0} & -x' \mathbf{X}^\top \\ \mathbf{0} & \mathbf{X}^\top & -y' \mathbf{X}^\top \end{bmatrix} \begin{bmatrix} \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix} = \mathbf{0}$$

How do we proceed?

$$\mathbf{p}_2^\top \mathbf{X} - \mathbf{p}_3^\top \mathbf{X} y' = 0$$

$$\mathbf{p}_1^\top \mathbf{X} - \mathbf{p}_3^\top \mathbf{X} x' = 0$$

In matrix form ...

$$\begin{bmatrix} \mathbf{X}^\top & \mathbf{0} & -x' \mathbf{X}^\top \\ \mathbf{0} & \mathbf{X}^\top & -y' \mathbf{X}^\top \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \mathbf{0}$$

For N points ...

$$\begin{bmatrix} \mathbf{X}_1^\top & \mathbf{0} & -x' \mathbf{X}_1^\top \\ \mathbf{0} & \mathbf{X}_1^\top & -y' \mathbf{X}_1^\top \\ \vdots & \vdots & \vdots \\ \mathbf{X}_N^\top & \mathbf{0} & -x' \mathbf{X}_N^\top \\ \mathbf{0} & \mathbf{X}_N^\top & -y' \mathbf{X}_N^\top \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \mathbf{0}$$

*How do we solve
this system?*

Solve for camera matrix by

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{A}\mathbf{x}\|^2 \text{ subject to } \|\mathbf{x}\|^2 = 1$$

$$\mathbf{A} = \begin{bmatrix} \mathbf{X}_1^\top & \mathbf{0} & -x' \mathbf{X}_1^\top \\ \mathbf{0} & \mathbf{X}_1^\top & -y' \mathbf{X}_1^\top \\ \vdots & \vdots & \vdots \\ \mathbf{X}_N^\top & \mathbf{0} & -x' \mathbf{X}_N^\top \\ \mathbf{0} & \mathbf{X}_N^\top & -y' \mathbf{X}_N^\top \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

Solution $\mathbf{x}$ is the column of $\mathbf{V}$
corresponding to smallest singular
value of

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

Almost there ...

$$\mathbf{P} = \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix}$$

How do you get the intrinsic and extrinsic parameters from the projection matrix?

Decomposition of the Camera Matrix

$$\mathbf{P} = \left[\begin{array}{ccc|c} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{array} \right]$$

$$\begin{aligned}\mathbf{P} &= \mathbf{K}[\mathbf{R}|\mathbf{t}] \\ &= \mathbf{K}[\mathbf{R} | -\mathbf{R}\mathbf{c}] \\ &= [\mathbf{M} | -\mathbf{M}\mathbf{c}]\end{aligned}$$

Find the camera center $\mathbf{c}$

$$\mathbf{P}\mathbf{c} = \mathbf{0}$$

SVD of $\mathbf{P}$!

$\mathbf{c}$ is the singular vector corresponding to the smallest singular value

Find intrinsic $\mathbf{K}$ and rotation $\mathbf{R}$

$$\mathbf{M} = \mathbf{K}\mathbf{R}$$


right upper triangle orthogonal

How do we find K and R ?

Decomposition of the Camera Matrix

$$\mathbf{P} = \left[\begin{array}{ccc|c} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{array} \right]$$

$$\begin{aligned}\mathbf{P} &= \mathbf{K}[\mathbf{R}|\mathbf{t}] \\ &= \mathbf{K}[\mathbf{R} | -\mathbf{R}\mathbf{c}] \\ &= [\mathbf{M} | -\mathbf{M}\mathbf{c}]\end{aligned}$$

Find the camera center $\mathbf{C}$

$$\mathbf{P}\mathbf{c} = \mathbf{0}$$

SVD of $\mathbf{P}$!

$\mathbf{c}$ is the singular vector corresponding
to the smallest singular value

Find intrinsic $\mathbf{K}$ and rotation $\mathbf{R}$

$$\mathbf{M} = \mathbf{K}\mathbf{R}$$

RQ decomposition

Geometric camera calibration

Given a set of matched points

$$\{\mathbf{X}_i, \mathbf{x}_i\}$$

point in 3D
space

point in the
image

*Where do we get these
matched points from?*

and camera model

$$\mathbf{x} = \mathbf{f}(\mathbf{X}; \mathbf{p}) = \mathbf{P}\mathbf{X}$$

projection
model

parameters

Camera
matrix

Find the (pose) estimate of

P

We'll use a **perspective** camera
model for pose estimation

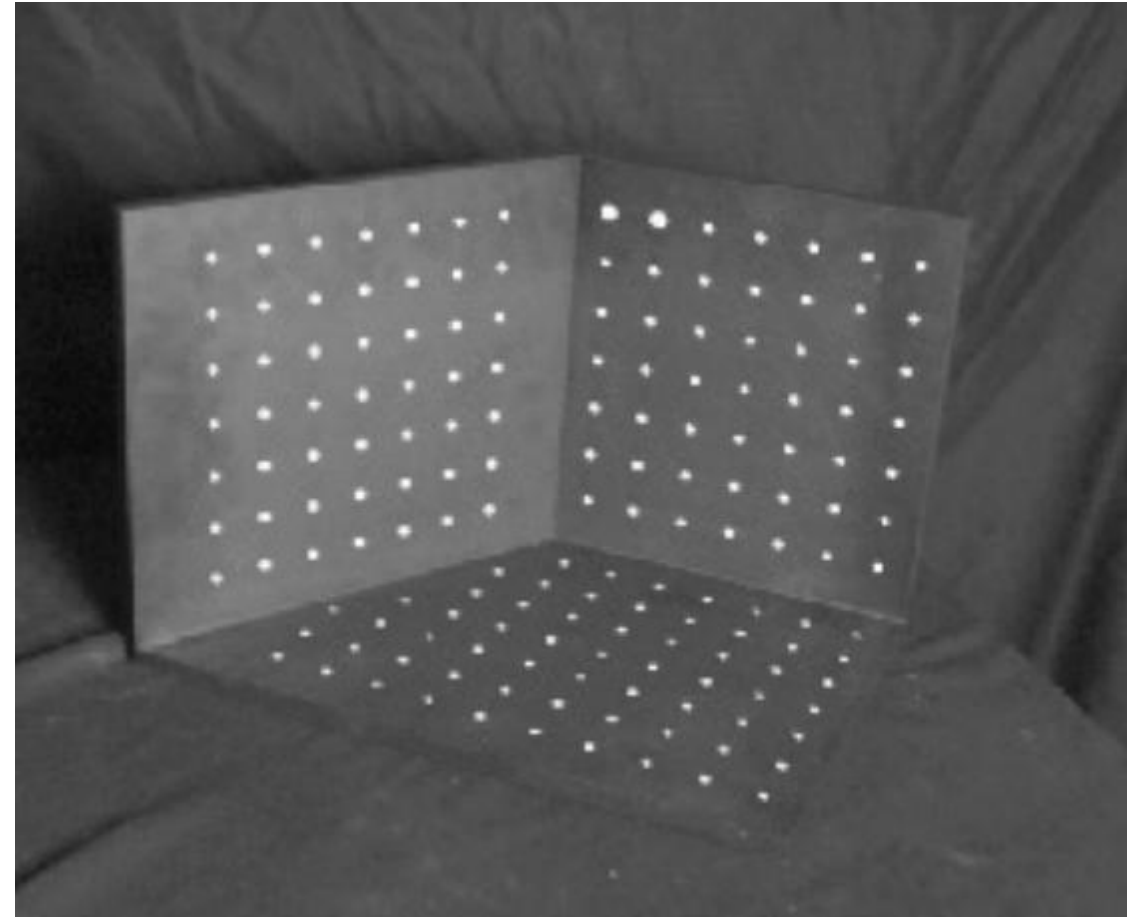
Calibration using a reference object

Place a known object in the scene:

- identify correspondences between image and scene
- compute mapping from scene to image

Issues:

- must know geometry very accurately
- must know 3D->2D correspondence



Geometric camera calibration

Advantages:

- Very simple to formulate.
- Analytical solution.

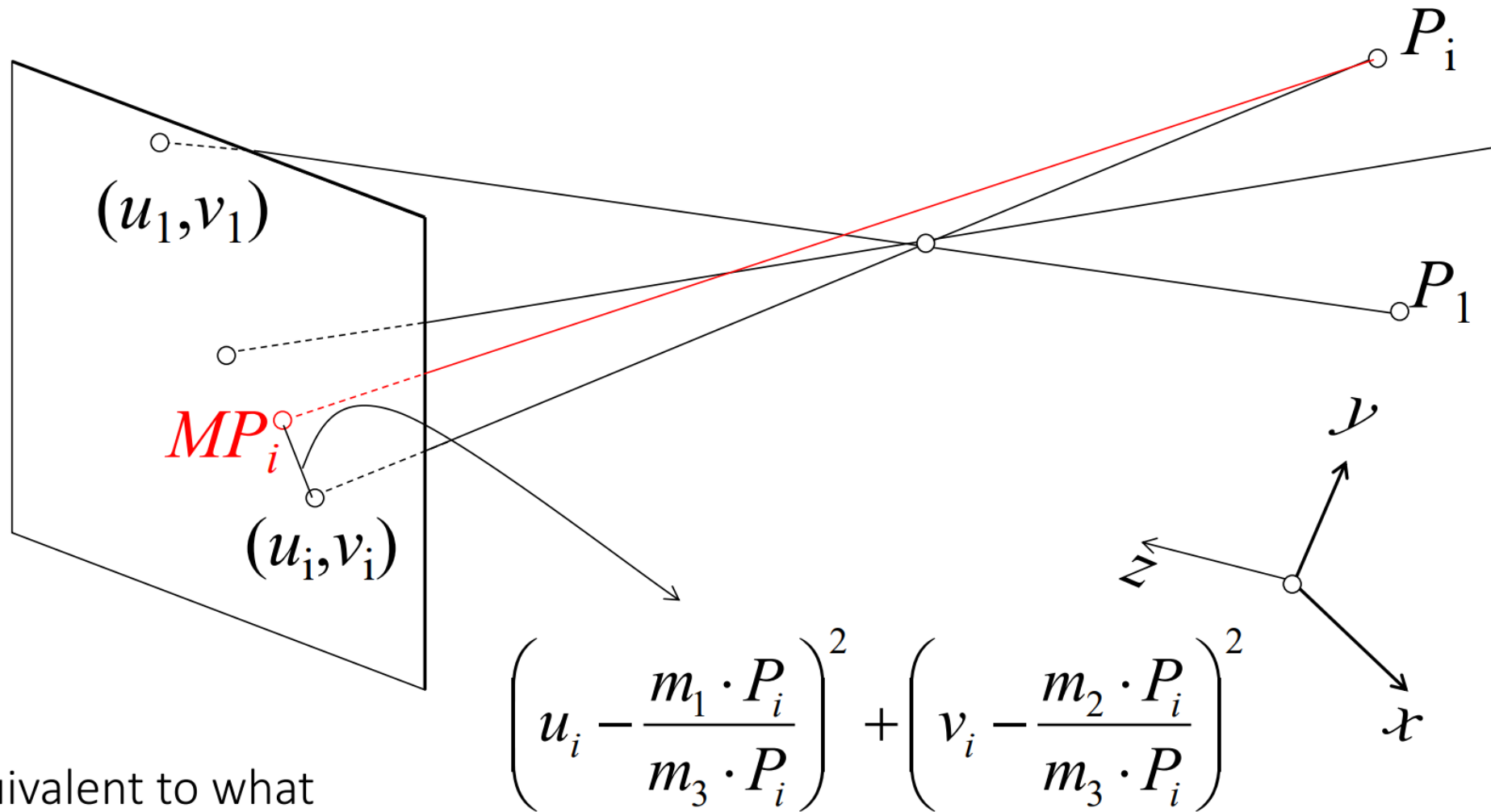
Disadvantages:

- Doesn't model radial distortion.
- Hard to impose constraints (e.g., known f).
- Doesn't minimize the correct error function.

For these reasons, *nonlinear methods* are preferred

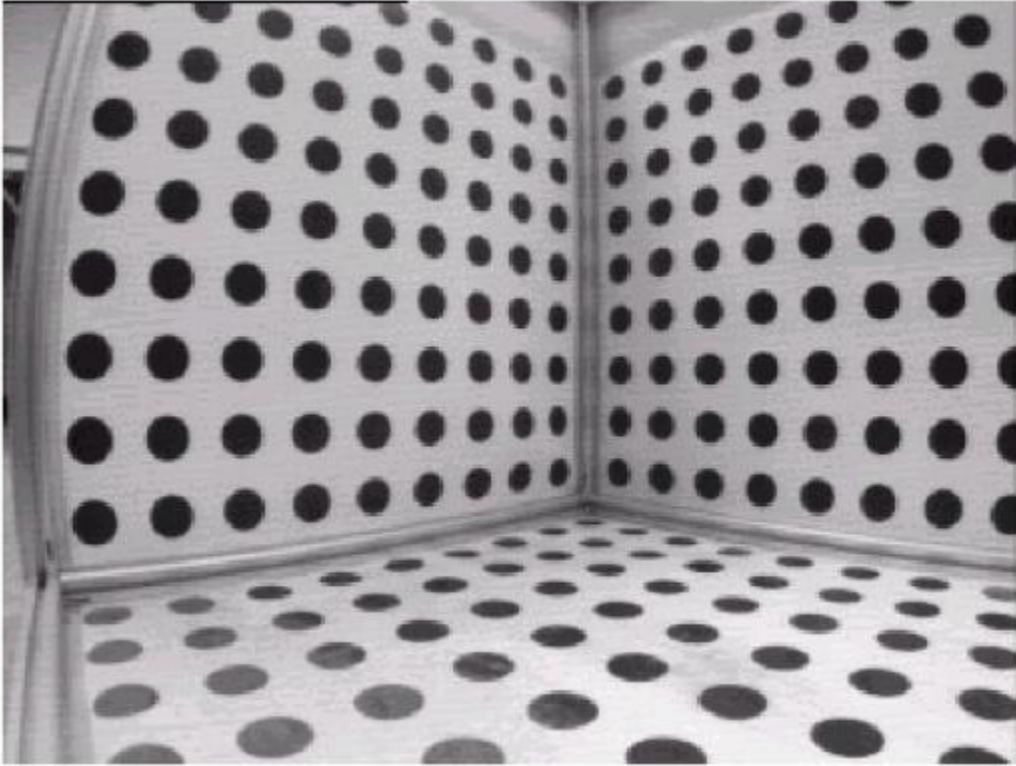
- Define error function E between projected 3D points and image positions
 - E is nonlinear function of intrinsics, extrinsics, radial distortion
- Minimize E using nonlinear optimization techniques

Minimizing reprojection error

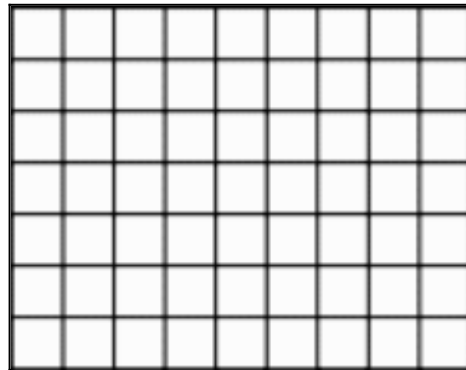


Is this equivalent to what we were doing previously?

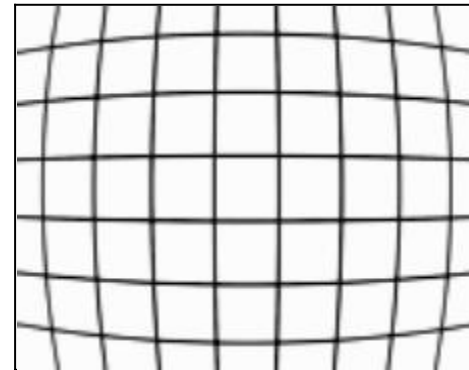
Radial distortion



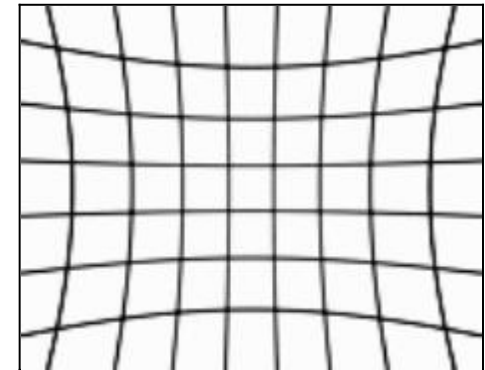
What causes this distortion?



no distortion

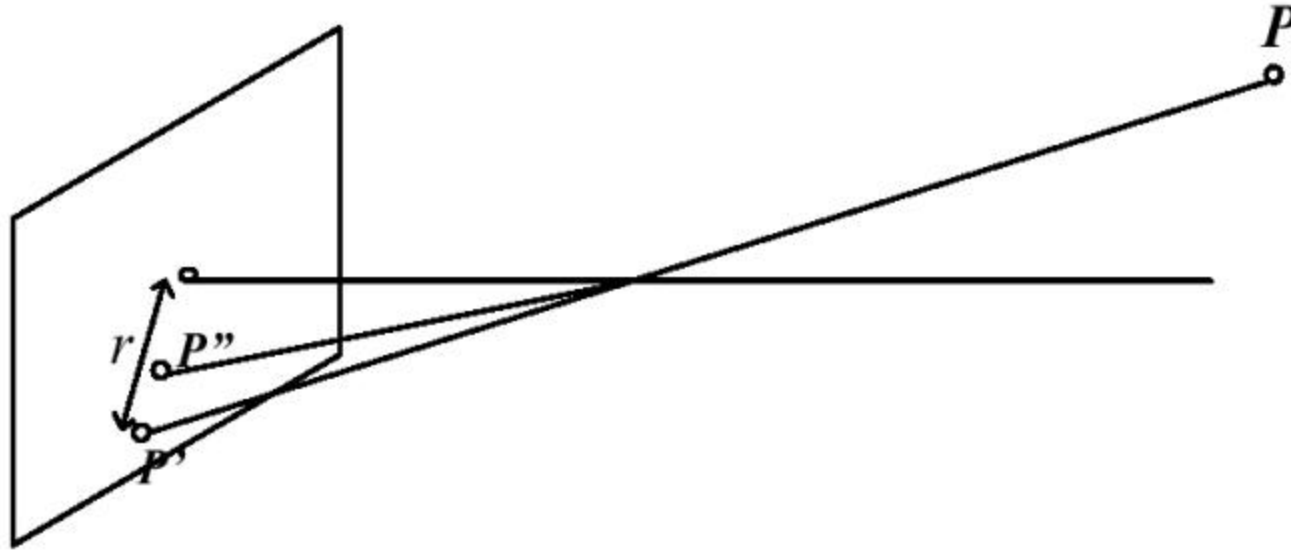


barrel distortion



pincushion distortion

Radial distortion model



Ideal:

$$x' = f \frac{x}{z}$$

$$y' = f \frac{y}{z}$$

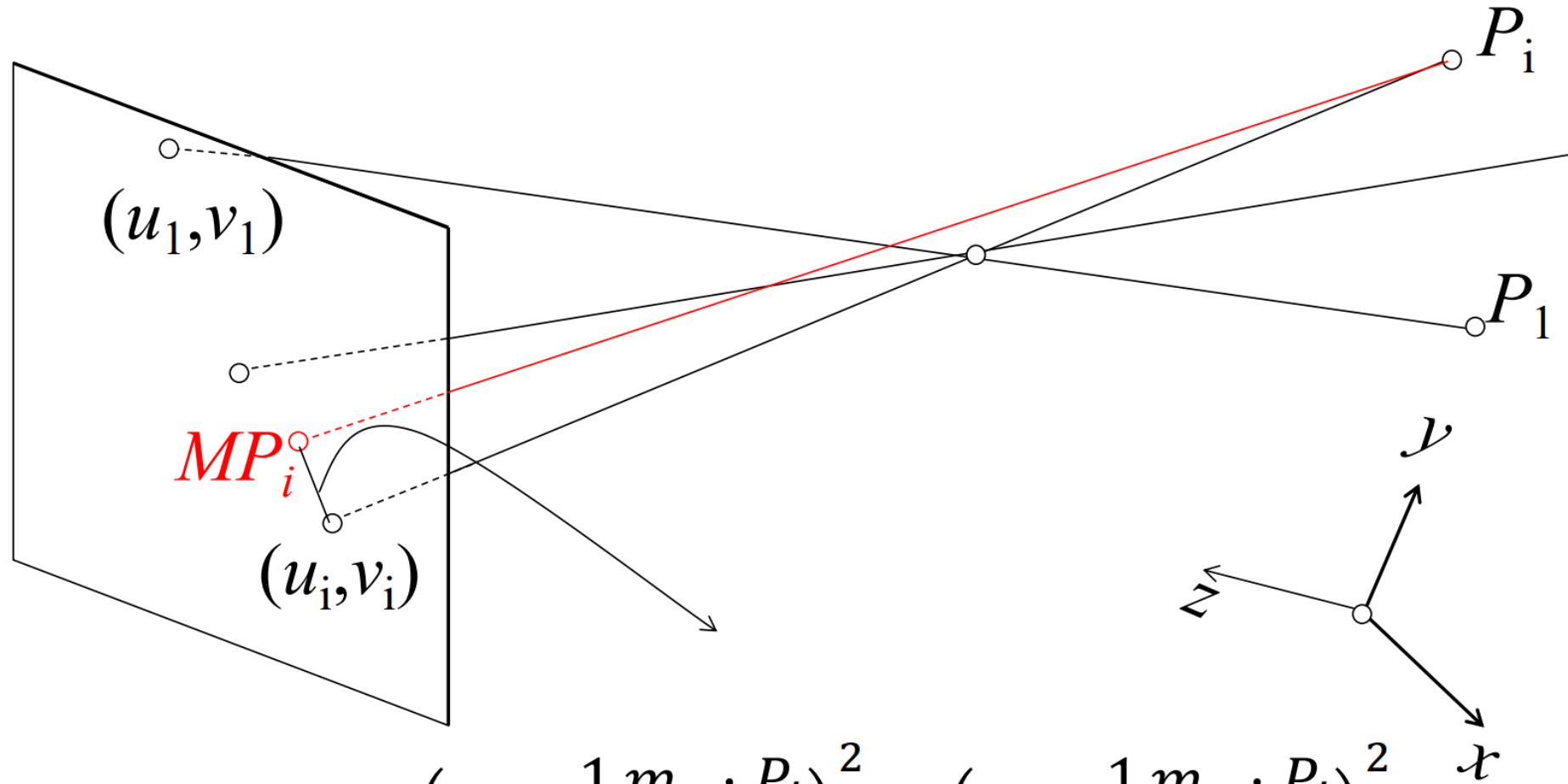
Distorted:

$$x'' = \frac{1}{\lambda} x'$$

$$y'' = \frac{1}{\lambda} y'$$

$$\lambda = 1 + k_1 r^2 + k_2 r^4 + \dots$$

Minimizing reprojection error with radial distortion



Add distortions to
reprojection error:

$$\left(u_i - \frac{1}{\lambda} \frac{m_1 \cdot P_i}{m_3 \cdot P_i}\right)^2 + \left(v_i - \frac{1}{\lambda} \frac{m_2 \cdot P_i}{m_3 \cdot P_i}\right)^2$$

Correcting radial distortion

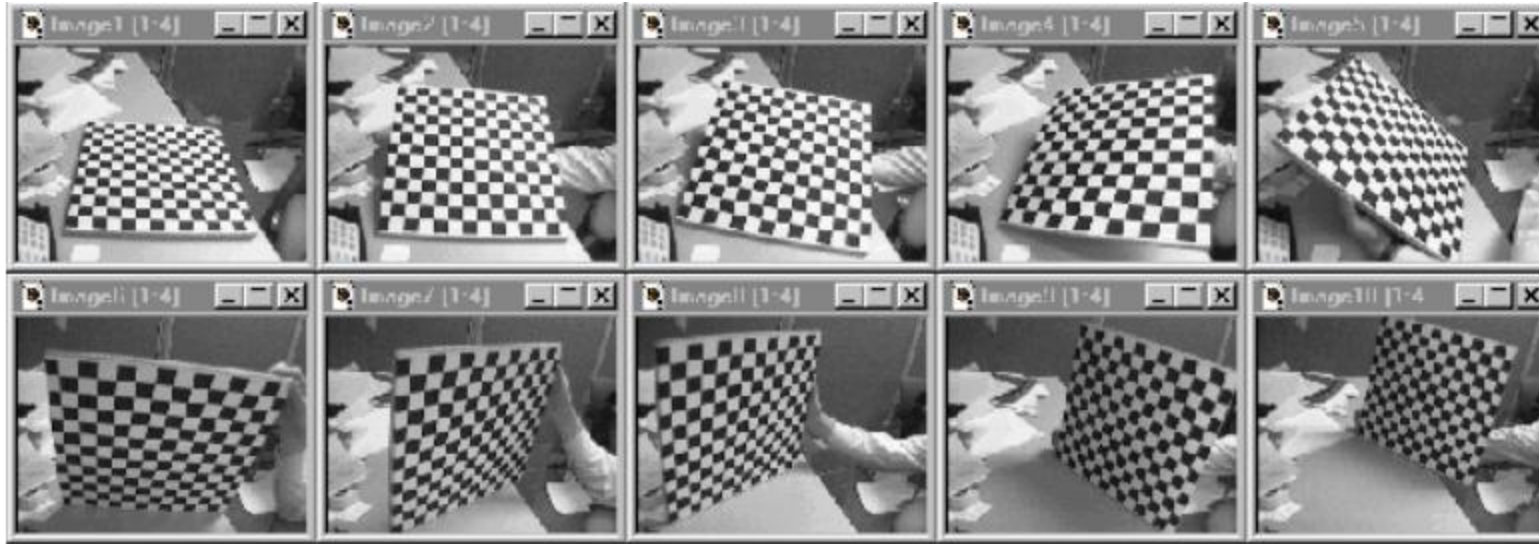


before



after

Alternative: Multi-plane calibration



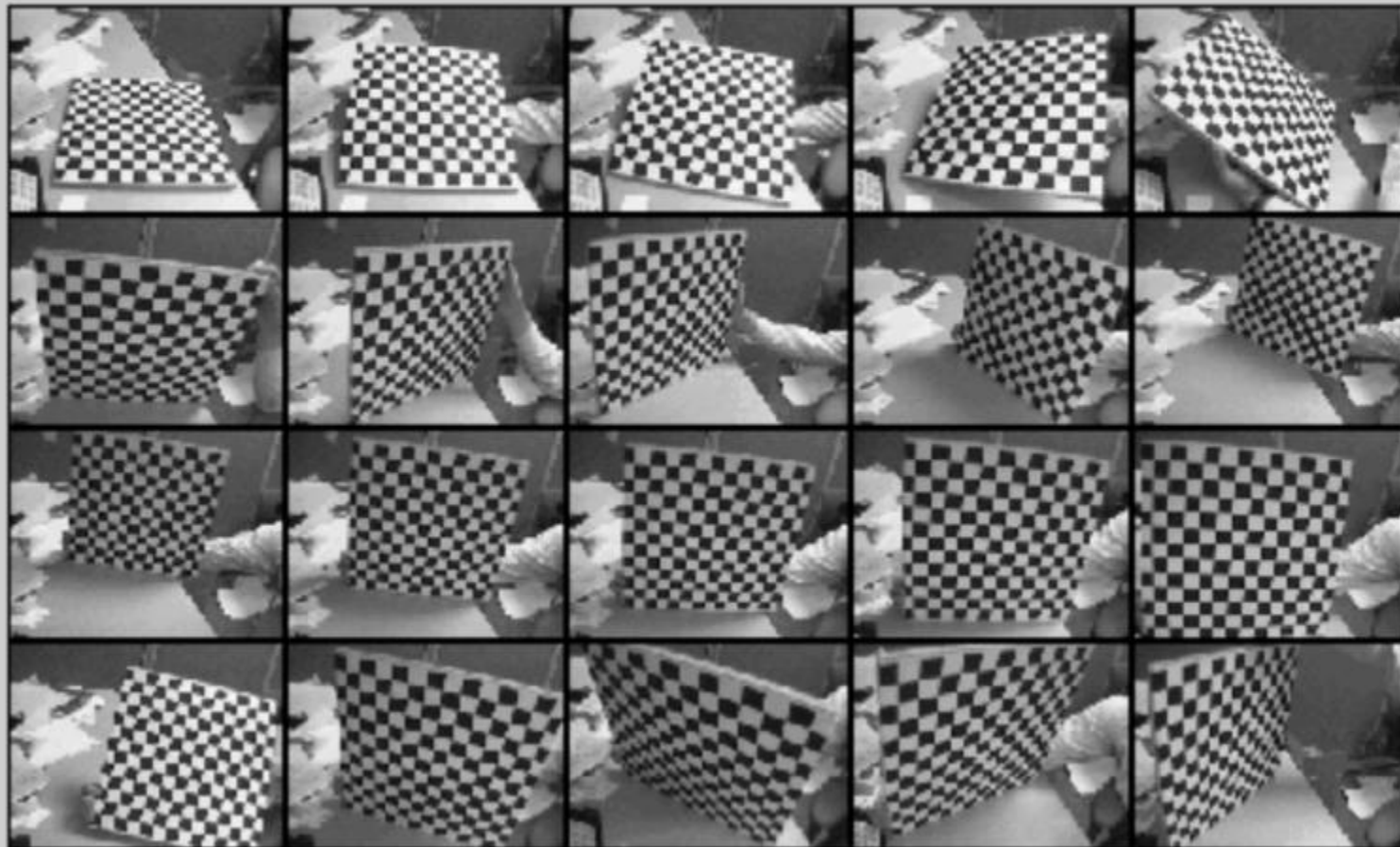
Advantages:

- Only requires a plane
- Don't have to know positions/orientations
- Great code available online!
 - Matlab version: http://www.vision.caltech.edu/bouguetj/calib_doc/index.html
 - Also available on OpenCV.

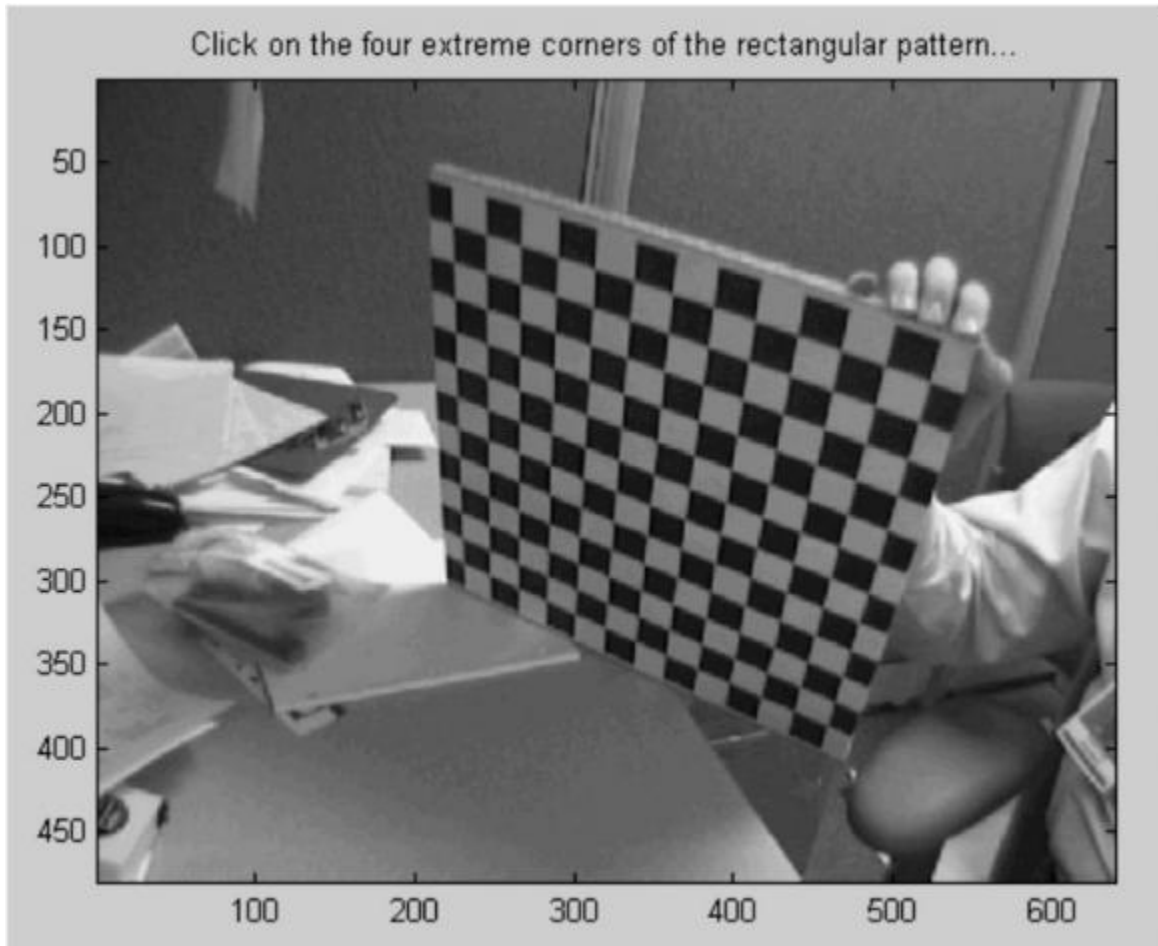
Disadvantage: Need to solve non-linear optimization problem.

Step-by-step demonstration

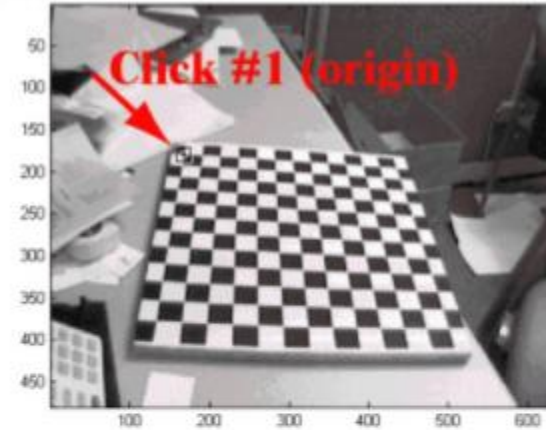
Calibration images



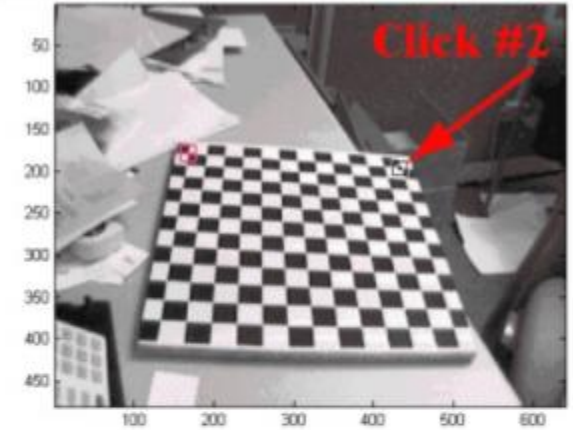
Step-by-step demonstration



Click on the four extreme corners of the rectangular pattern (first corner = origin) ... Image 1



Click on the four extreme corners of the rectangular pattern (first corner = origin) ... Image 1



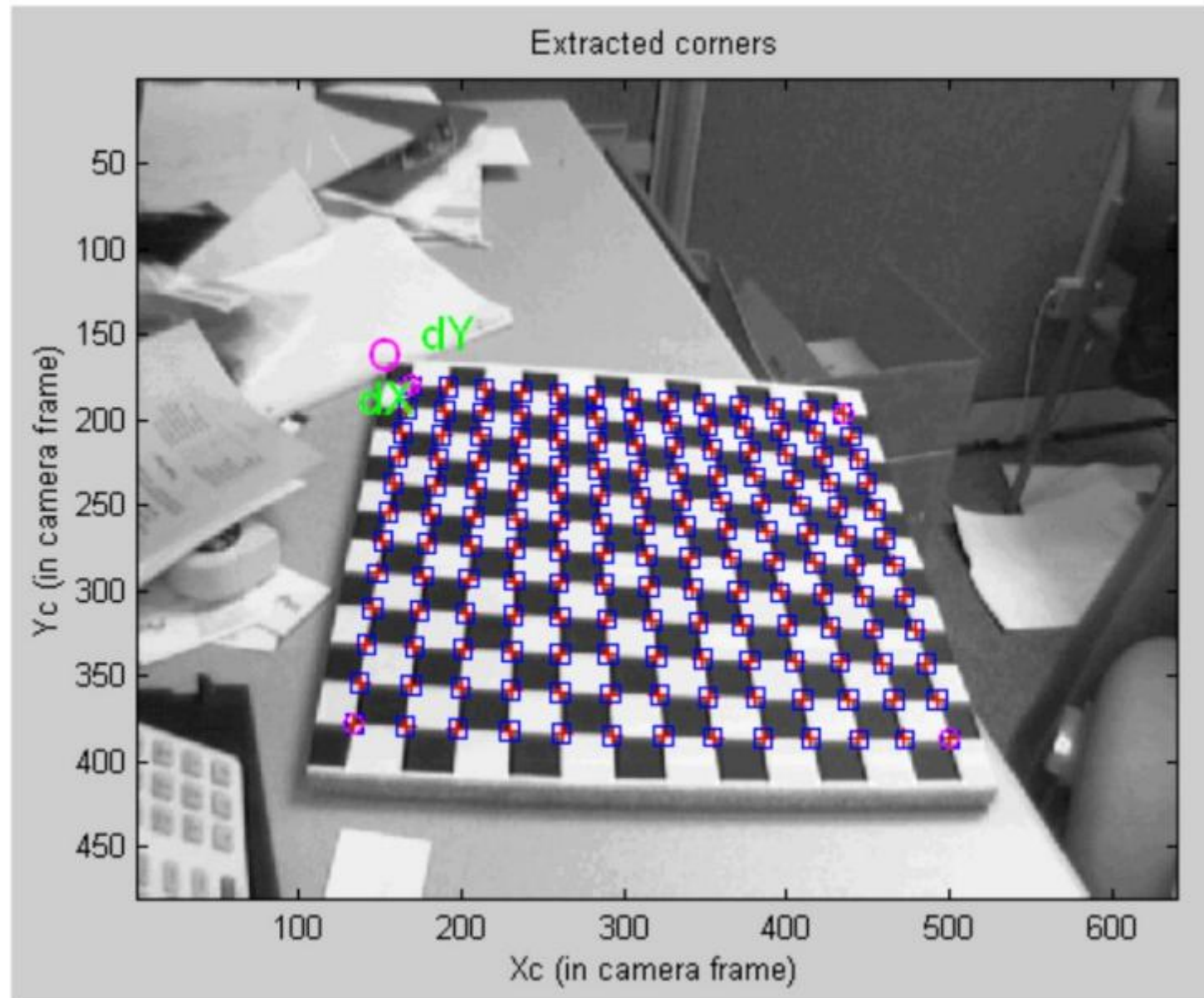
Click on the four extreme corners of the rectangular pattern (first corner = origin) ... Image 1



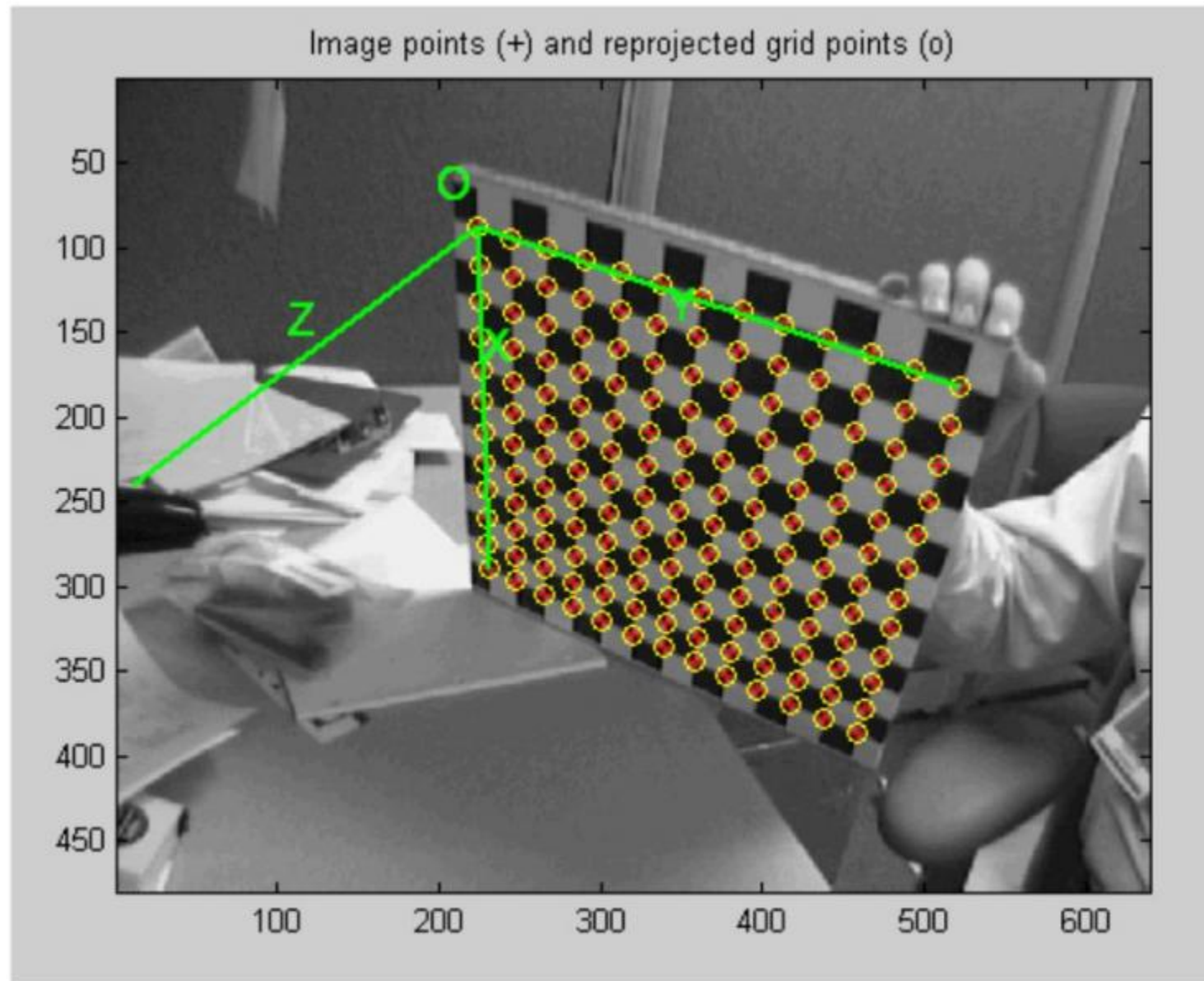
Click on the four extreme corners of the rectangular pattern (first corner = origin) ... Image 1



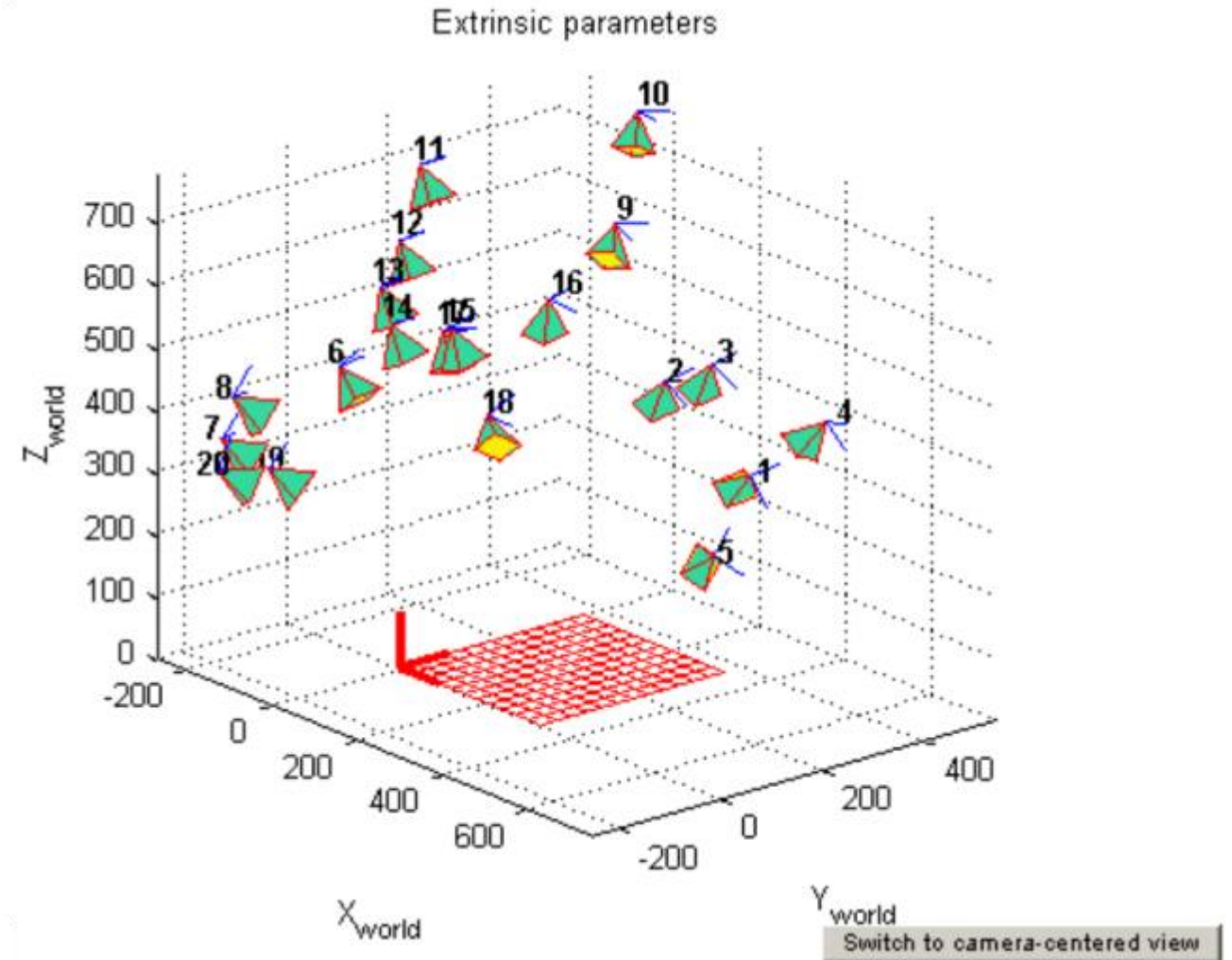
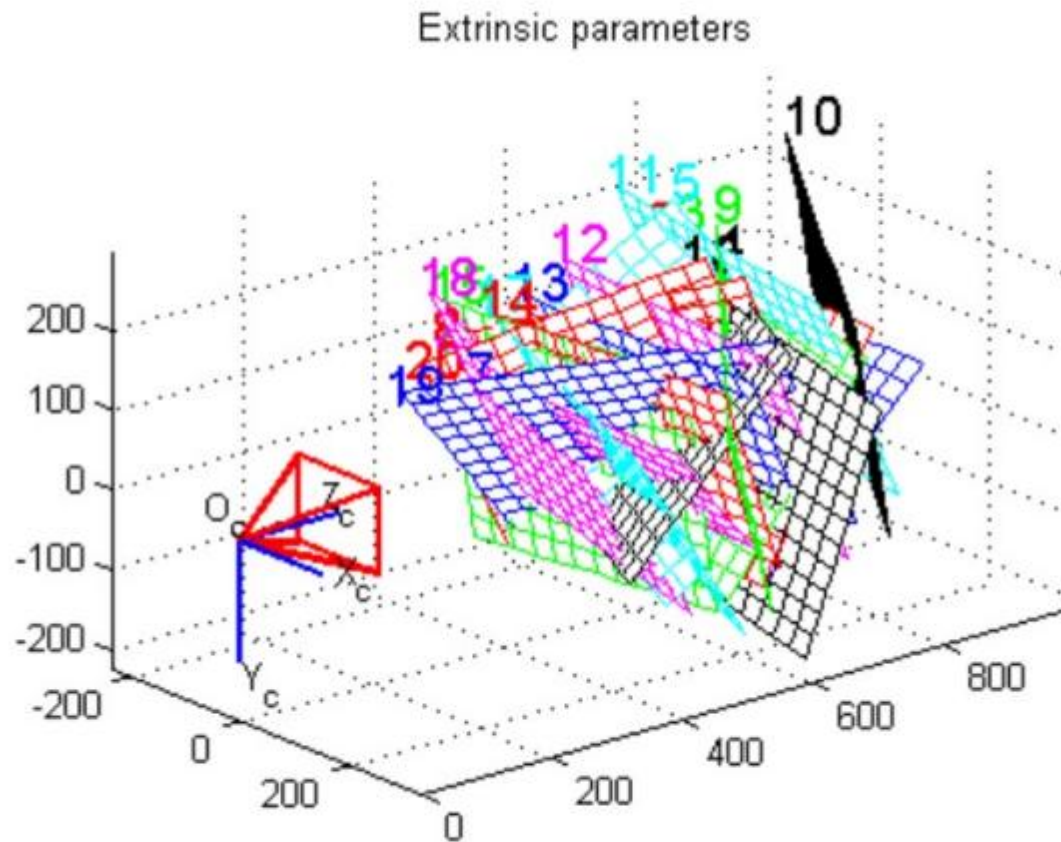
Step-by-step demonstration



Step-by-step demonstration



Step-by-step demonstration

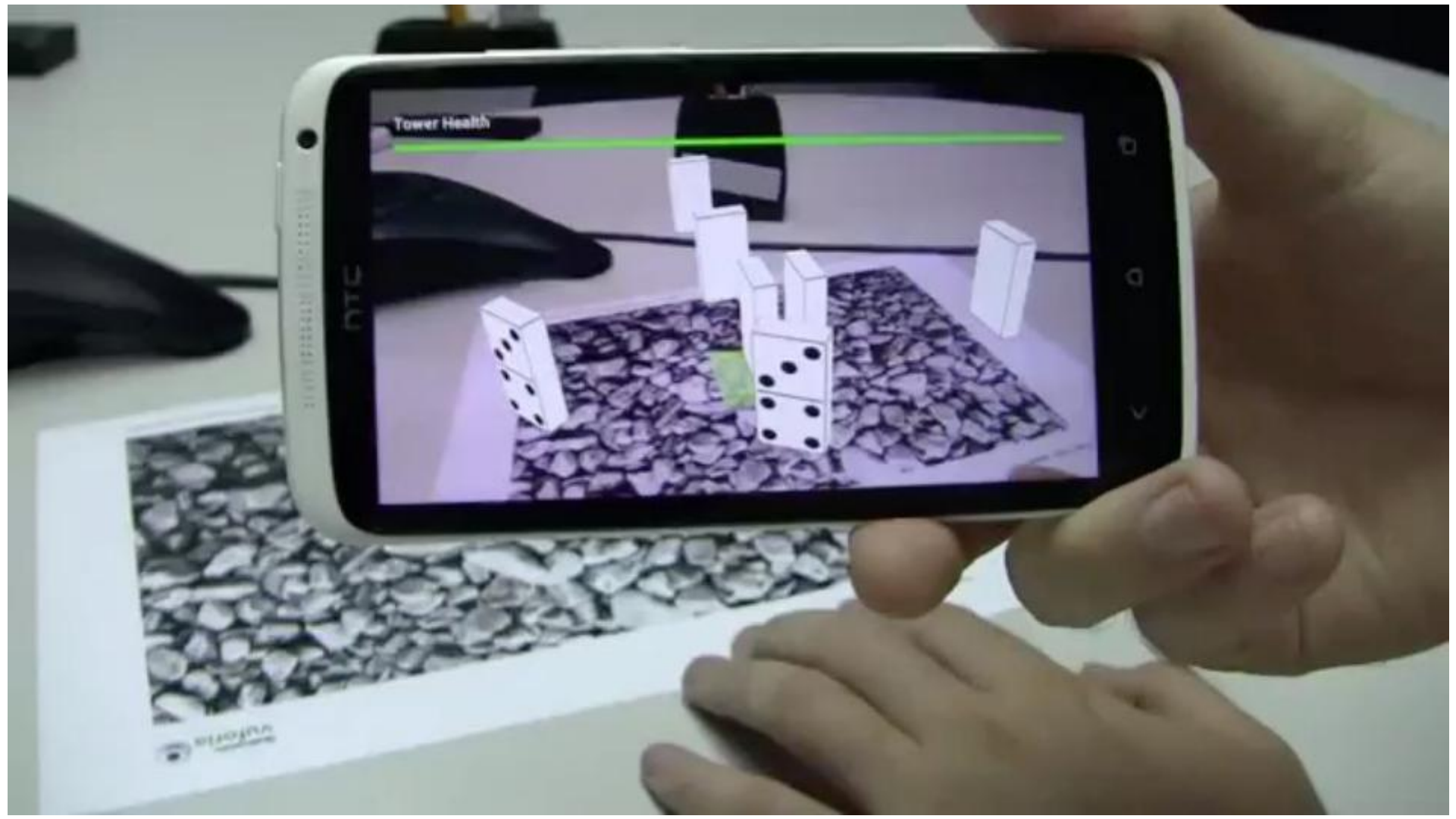


What does it mean to “calibrate a camera”?

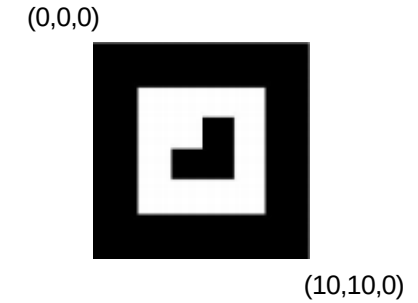
What does it mean to “calibrate a camera”?

Many different ways to calibrate a camera:

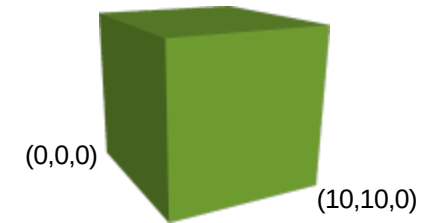
- Radiometric calibration.
- Color calibration.
- Geometric calibration.
- Noise calibration.
- Lens (or aberration) calibration.



3D locations of planar marker features are known in advance



3D content prepared in advance



Simple AR program

1. Compute point correspondences (2D and AR tag)
2. Estimate the pose of the camera **P**
3. Project 3D content to image plane using **P**

本学期所学习的主要内容

基本方法考察

✓ 评估方法

- 类别不平衡条件下的评估：TP, TN, FP, FN, ROC和PR曲线, F1 score

✓ Fisher线性判别分析

- 类内散度, 类间散度, Fisher准则函数, 多分类问题

✓ 概率方法

- 参数化估计, 非参数化估计

✓ 度量方法

- 度量的定义, L_p 度量, 度量学习

本学期所学习的主要内容

✓ 隐马尔可夫模型HMM

- 两个基本假设, 三大核心问题

是什么

✓ 图像滤波

- 空域与频域方法, 振铃效应 (aliasing), 奈奎斯特-香农采样定理

✓ Harris角点检测

- 方法流程, 结构张量, 响应函数, 性质

✓ 图像变换, 相机模型

- 各类变换形式与自由度, 单应变换求解,

评价不平衡时的准则(1)

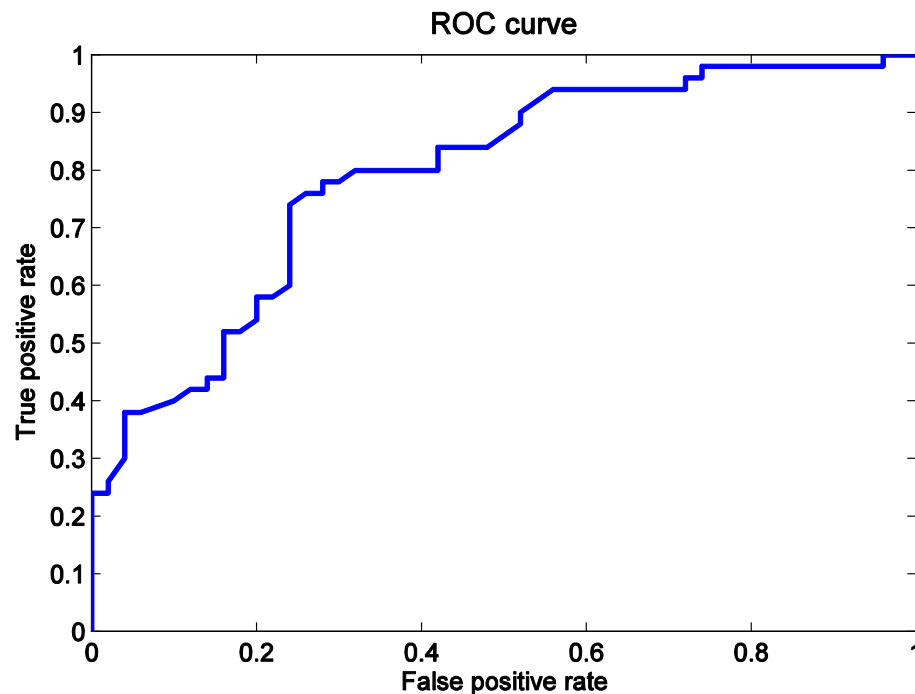
	预测为positive	预测为negative
真实值为positive	True positive (真阳性)	False negative(伪阴性)
真实值为negative	False positive(伪阳性)	True negative(真阴性)

- ✓ TP、TN、FP、FN：标记四种情况的样例数目
- ✓ TOTAL：总数 $TP+TN+FP+FN$
 - 正样本数目： $P = TP+FN$ ，负样本数目： $N = FP+TN$
- ✓ False positive rate: $FPR = FP / N$
- ✓ False negative rate: $FNR = FN / P$
- ✓ True positive rate: $TPR = TP / P$
- ✓ Accuracy: $ACC = (TP+TN) / TOTAL$

评价不平衡时的准则(2)

✓ AUC-ROC (Area Under the ROC Curve)

- ROC – Receiver operating characteristic



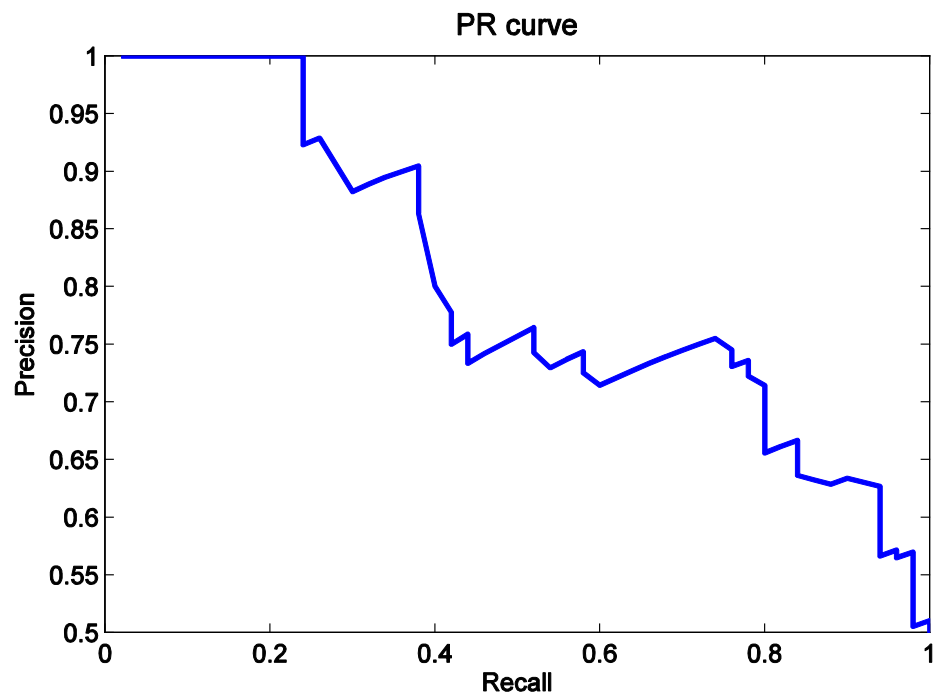
- Y轴: TPR
- X轴: FPR
- 其值为面积
- 为什么?
- 对角线是?
- 非减

评价不平衡时的准则(3)

- ✓ Precision（查准率）： $PRE = TP / (TP + FP)$
- ✓ Recall（查全率）： $REC = TP / P$ (和TPR一样)
- ✓ F1 score: Precision和Recall的调和平均(harmonic mean)
 - 调和平均： $\left(\frac{x^{-1} + y^{-1}}{2} \right)^{-1} = \frac{2xy}{x+y}$
 - $F1 = 2TP / (2TP + FP + FN)$
 - 为什么？

评价不平衡时的准则(4)

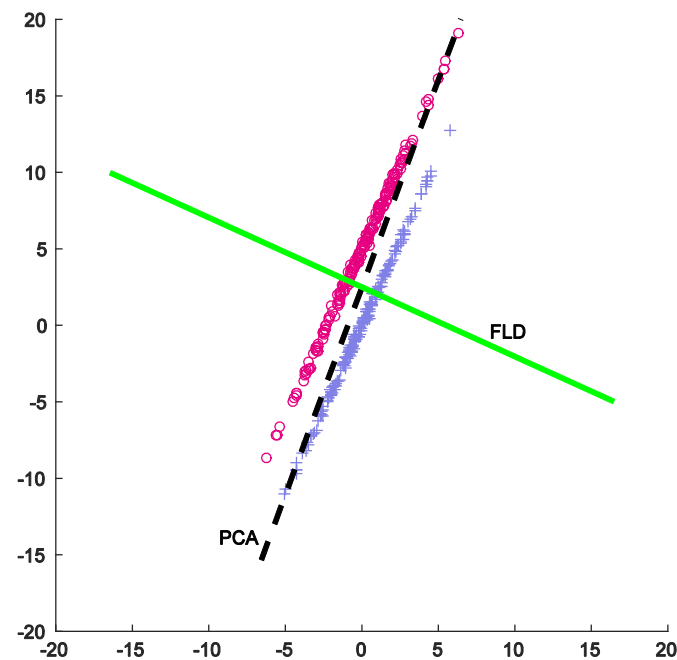
✓ AUC-PR (Area Under the Precision-Recall Curve)



- Y轴: Precision
- X轴: Recall
- 其值为面积
- 为什么?
- 单调吗?

进一步阅读: [The Relationship Between Precision-Recall and ROC Curves](#)

Idea: FLD的动机 (motivation)



Bad linear feature (PCA)

Good linear feature (FLD)

Objective: Fisher's Criterion

- ✓ 怎样描述“分开”的程度(separation)?
- ✓ Maximize $(m_2 - m_1)^2$? 问题?
 - 这个值可以无限大。怎么解决?
 - 加限制条件 $\mathbf{w}^T \mathbf{w} = 1$
 - 看前面的图，这个值不是越大越好。怎么解决?
- ✓ Fisher准则
 - 在要求 $|m_2 - m_1|$ 尽量大的同时，要求两类在投影以后尽量集中，或者不分散。怎么度量分散程度?

$$J(\mathbf{w}) = \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2}$$

散度矩阵

✓ $S_k = \sum_{y_i=k} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^T$ 是什么？

✓ 类内散度矩阵 within-class scatter matrix

$$S_W = S_1 + S_2$$

✓ 类间散度矩阵 between-class scatter matrix

$$S_B = (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T$$

✓ Fisher准则的矩阵形式

- $\max J(\mathbf{w}) = \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}, \text{ s.t. } \mathbf{w}^T \mathbf{w} = 1$

- 这种形式称为广义瑞利商 generalized Rayleigh quotient

Optimization: 如何求解？

- ✓ (Simplification/transformation) : 用拉格朗日乘子法，证明（记得查表）最优时必须满足

$$S_B \mathbf{w} = \lambda S_W \mathbf{w}$$

- ✓ 该问题称为广义特征值generalized eigenvalue问题
 - 得到“ S_B 和 S_W ”的广义特征值和广义特征向量
 - Generalized eigenvalue (eigenvector) of S_B and S_W
- ✓ 但是我们不用去解这个问题
 - $S_B \mathbf{w} = (\mu_2 - \mu_1)(\mu_2 - \mu_1)^T \mathbf{w} \propto (\mu_2 - \mu_1)$
 - $(\mu_2 - \mu_1) = \lambda S_W \mathbf{w}!$

如果大于2类怎么办？

✓ C 类问题

- μ_i, N_i, m_i, S_i 和2类问题中一样定义
- $S_W = \sum_{i=1}^C S_i$, 很容易从2类问题推广
- 定义 $N = \sum_{i=1}^C N_i$
- 定义总均值 $\mu = \frac{1}{N} \sum_{i=1}^C N_i \mu_i = \frac{1}{N} \sum_x x$

✓ S_B 没有定义，无法直接从2类问题推广

- 总散度矩阵total scatter matrix, $S_T = \sum_x (x - \mu)(x - \mu)^T$
- $S_T = S_W + \sum_{i=1}^C N_i (\mu_i - \mu)(\mu_i - \mu)^T = S_W + S_B$
- 定义多类的 $S_B = \sum_{i=1}^C N_i (\mu_i - \mu)(\mu_i - \mu)^T$
- 证明，当 $C = 2$ 时，有 $S_T = S_W + S_B$

更多的投影方向

$$\max J(\mathbf{w}) = \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}$$

- ✓ 求解广义特征值问题

$$S_B \mathbf{w}_i = \lambda_i S_W \mathbf{w}_i$$

- ✓ 最多能得到 $C - 1$ 个有效的投影方向
 - 为什么？

贝叶斯参数估计

✓ Bayesian parameter estimation

- MLE: 视 θ 为固定的参数, 假设存在一个最佳的参数 (或参数的真实值是存在的), 目的是找到这个值
 - MAP: 将 $p(\theta)$ 的影响代入MLE中, 仍然假设存在最优的参数
 - 以上均称为点估计point estimation
- ✓ 在贝叶斯观点中, θ 是一个分布/随机变量, 所以估计应该是估计一个分布, 而不是一个值 (点) !
- $p(\theta|D)$: 这是贝叶斯参数估计的输出, 是一个完整的分布, 而不是一个点

距离度量

- ✓ 距离是一种度量不相似度的指标
- ✓ 度量（metric）是一个函数
- ✓ 欧氏距离是最常用的距离度量：

- ✓ $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$

- ✓ 满足以下性质：

- ✓ i. $d(\mathbf{x}, \mathbf{y}) \geq 0$

非负性

- ✓ ii. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$

对称性

- ✓ iii. $d(\mathbf{x}, \mathbf{y}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{y}$

同一性

- ✓ iv. $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$

三角不等式

向量范数和度量

- ✓ 欧氏距离和向量范数的关系: $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$
- ✓ 只考虑实向量, 向量范数 f 满足三个性质:
 - ✓ i. $f(c\mathbf{x}) = |c|f(\mathbf{x})$ 齐次性
 - ✓ ii. $f(\mathbf{x}) = 0 \leftrightarrow \mathbf{x} = 0$ 非负性
 - ✓ iii. $f(\mathbf{x} + \mathbf{y}) \leq f(\mathbf{x}) + f(\mathbf{y})$ 三角不等式
- ✓ 容易证明 $f(\mathbf{0}) = 0, f(-\mathbf{x}) = f(\mathbf{x}), f(\mathbf{x}) \geq 0$
- ✓ 容易证明 $f(\mathbf{x} - \mathbf{y})$ 满足度量的所有条件
- ✓ 向量范数可以诱导出度量

l_p 范数和 l_p 度量

✓ l_p 范数在 $p \geq 1$ 时有定义:

✓
$$\|\mathbf{x}\|_p = (\sum_{i=1}^d |x_i|^p)^{\frac{1}{p}}$$

✓ 容易证明 l_p 范数满足齐次性和非负性

✓ 闵可夫斯基不等式

✓
$$(\sum_{i=1}^d (x_i + y_i)^p)^{\frac{1}{p}} \leq (\sum_{i=1}^d x_i^p)^{\frac{1}{p}} + (\sum_{i=1}^d y_i^p)^{\frac{1}{p}}$$

✓ L1 度量相比于 L2 度量对噪声更鲁棒: L1 损失线性增长, L2 损失二次增长。

距离度量学习

✓ 马氏距离

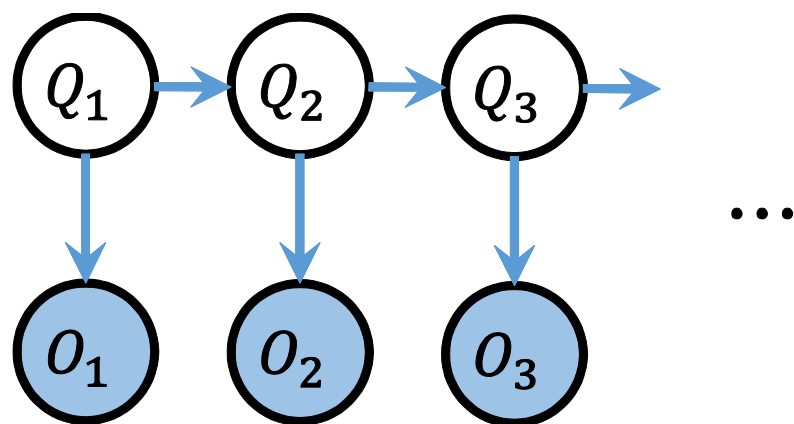
- 令 G 为任意 $d \times d$ 的正定矩阵，那么
- $f(\mathbf{x}) = \|G\mathbf{x}\|$ 定义了一个有效的向量范数
- 对应的距离度量为 $\|G(\mathbf{x} - \mathbf{y})\|$
- 常用平方距离：
- $d_A^2(\mathbf{x}, \mathbf{y}) = (\mathbf{x} - \mathbf{y})^T A (\mathbf{x} - \mathbf{y})$, $A = G^T G$

✓ 为什么要引入马氏距离？

- 假设数据服从高斯分布
- 白化变换 $\Sigma^{-\frac{1}{2}}(\mathbf{x} - \bar{\mathbf{x}})$

怎样在模式识别中发挥更大作用？

- ✓ 例如，在连续手写识别中，笔画stroke没有在DTMC里面用到，那么DTMC就没法用于连续手写识别
 - 想办法把笔画加进去！
 - 状态是什么？



HMM中要解决的问题

- ✓ 怎样设计状态？ -- 自动学习？
- ✓ 怎样设计观察值？ -- 根据问题的特点和实践反复设计
- ✓ 与具体问题无关的
 - 指定一个HMM需要的所有参数： $\lambda = (\pi, A, B)$
 - 问题1： Evaluation估值
 - 问题2： Decoding解码
 - 问题3： Learning学习

Problem 1. Evaluation

✓ 输入

- 一个完全指定的HMM模型，即 $\lambda = (\boldsymbol{\pi}, A, B)$ 已知
- 一个完全观测的输出序列 $O_1 O_2 \cdots O_T$ ，或 $\boldsymbol{O} = O_{1:T}$

✓ 输出

- $P(\boldsymbol{O}|\lambda)$ – 含义是？
- 在这个模型 λ 中观察到特定输出 $\boldsymbol{O}$ 的概率

✓ 作用是？

- 可以看出此模型对该观察序列的成绩score
- 可以用来从多个模型中选择最适合的模型

Problem 2: Decoding

✓ 输入

- 一个完全指定的HMM模型，即 $\lambda = (\pi, A, B)$ 已知
- 一个完全观测的输出序列 $O_1 O_2 \cdots O_T$ ，或 $\mathbf{O} = O_{1:T}$
- 某个标准criterion

✓ 输出

- 一个完全指定的隐变量序列 $X_{1:T}$ 的值

✓ 作用是？

- 如，语音识别中状态可能有实际意义（各音节）
 - 唯一吗？
- 可以用来观察模型结构，优化模型

Problem 3: Learning

✓ 输入

- 网络结构，状态数、输出数
- 若干观测序列 $\{\mathbf{O}\}$

✓ 输出

- 最优的参数 $\lambda = (\boldsymbol{\pi}, A, B)$ 使得 $P(\{\mathbf{O}\}|\lambda)$ 最大

✓ 作用

- 显而易见
- 最重要的问题
- 有时候一个足够长的观测序列就够了

Convolution for 2D discrete signals

Definition of filtering as convolution:

$$(f * I)(x, y) = \sum_{i, j = -\infty}^{\infty} f(i, j) I(x - i, y - j)$$

Diagram illustrating the convolution operation with annotations:

- filtered image**: points to $(f * I)(x, y)$
- filter**: points to $f(i, j)$
- input image**: points to $I(x - i, y - j)$
- notice the flip**: points to the minus signs in $x - i$ and $y - j$

If the filter $f(i, j)$ is non-zero only within $-1 \leq i, j \leq 1$,
then

$$(f * I)(x, y) = \sum_{i, j = -1}^1 f(i, j) I(x - i, y - j)$$

The kernel we saw earlier is the 3x3 matrix representation of $f(i, j)$.

The convolution theorem

The Fourier transform of the convolution of two functions is the product of their Fourier transforms:

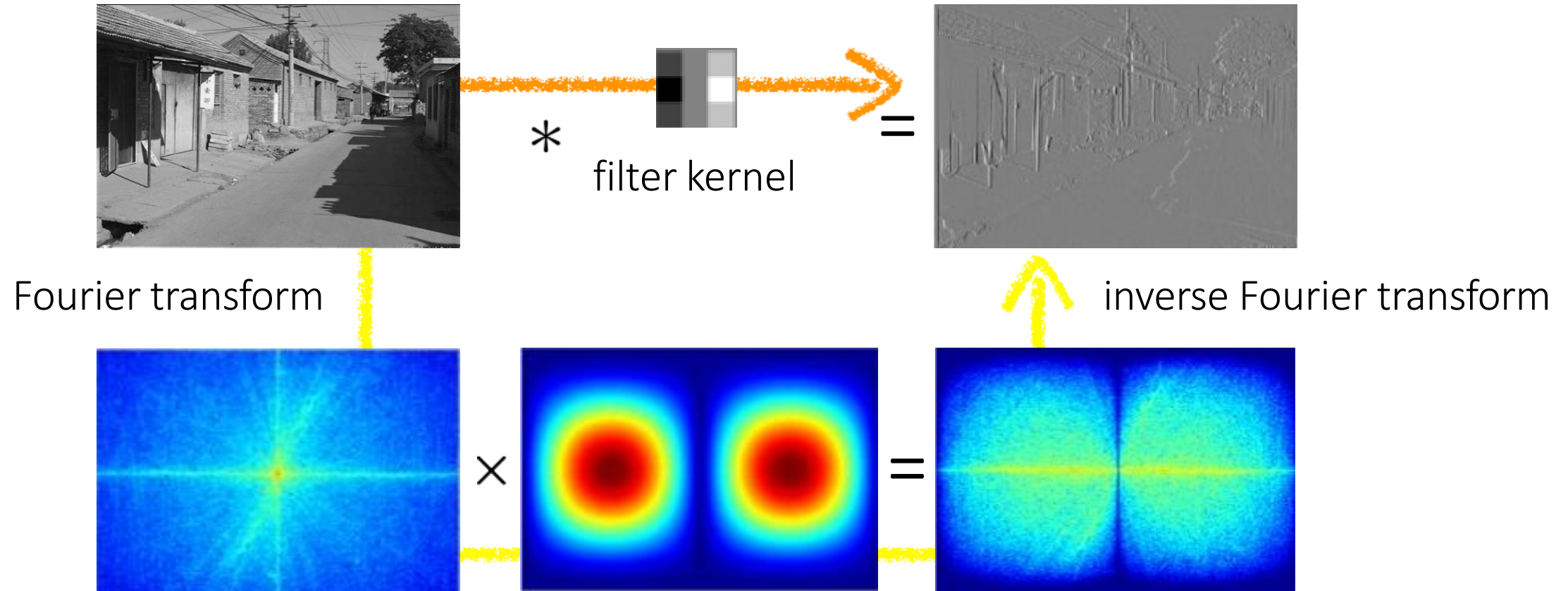
$$\mathcal{F}\{g * h\} = \mathcal{F}\{g\}\mathcal{F}\{h\}$$

The inverse Fourier transform of the product of two Fourier transforms is the convolution of the two inverse Fourier transforms:

$$\mathcal{F}^{-1}\{gh\} = \mathcal{F}^{-1}\{g\} * \mathcal{F}^{-1}\{h\}$$

Convolution in spatial domain is equivalent to multiplication in frequency domain!

Spatial domain filtering

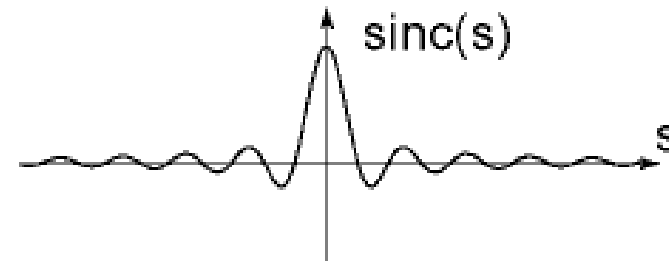
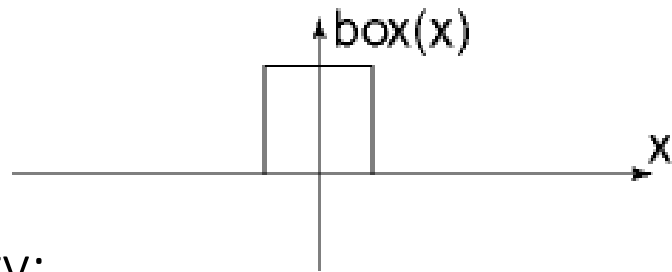


Frequency domain filtering

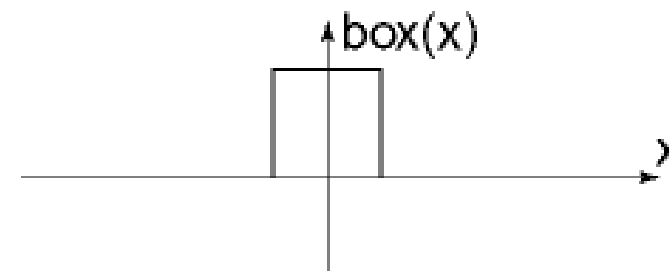
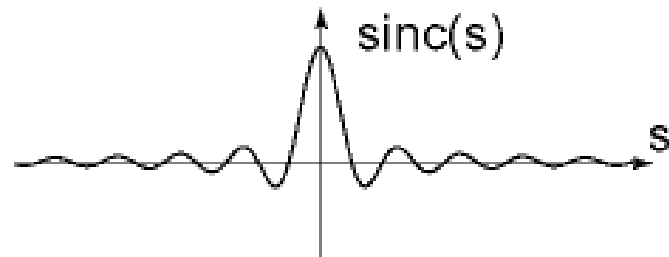
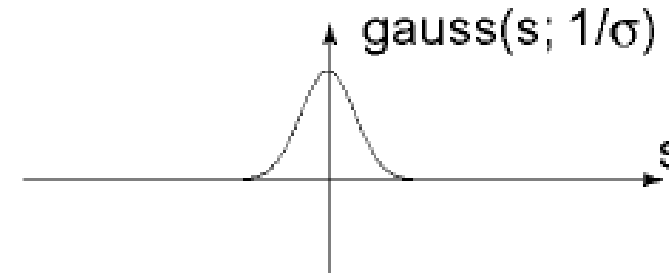
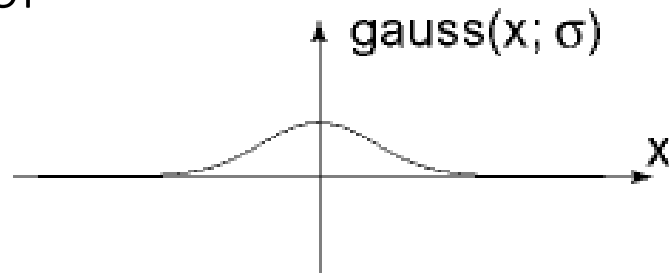
Fourier transform pairs

spatial domain

frequency domain

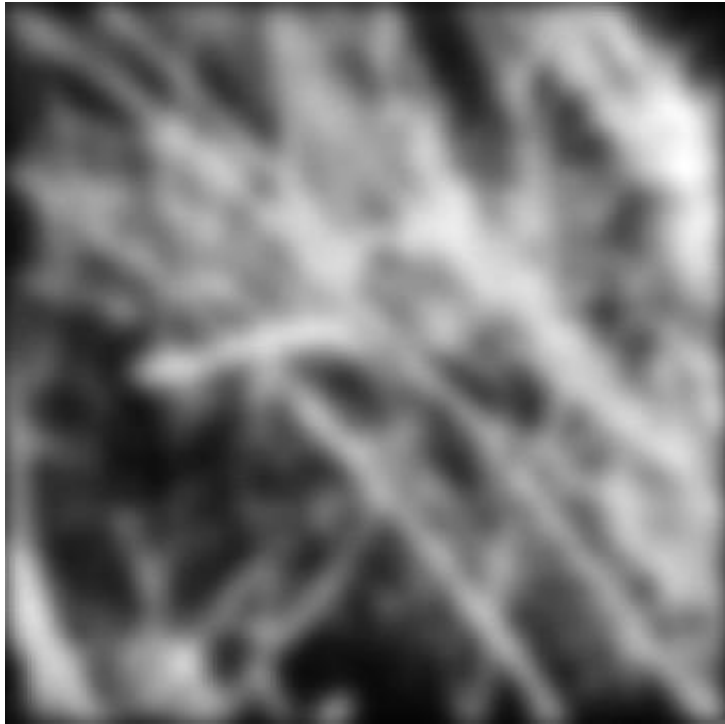


Note the symmetry:
duality property of
Fourier transform

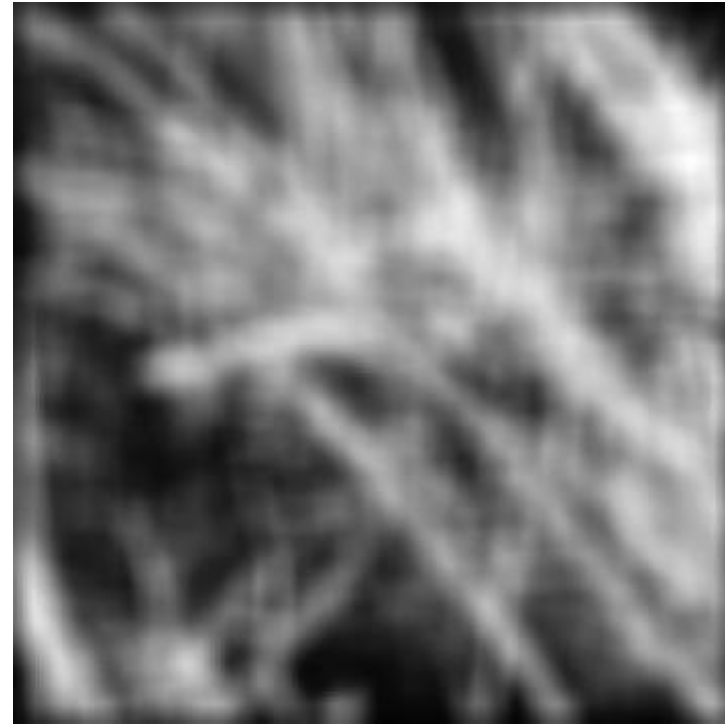
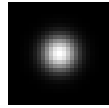


Revisiting blurring

Why does the Gaussian give a nice smooth image, but the square filter give edgy artifacts?



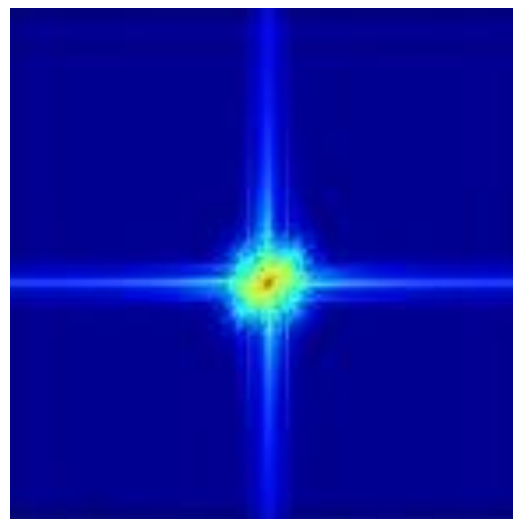
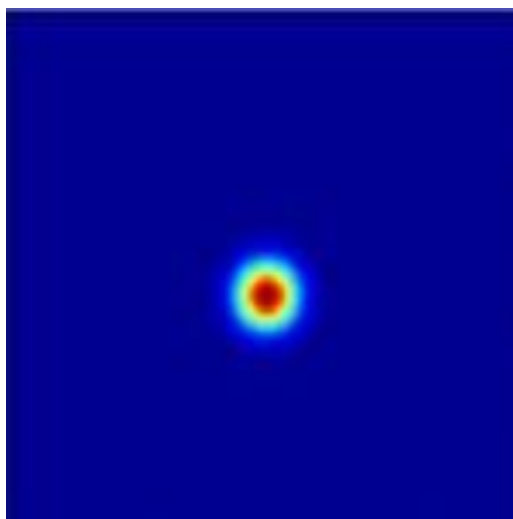
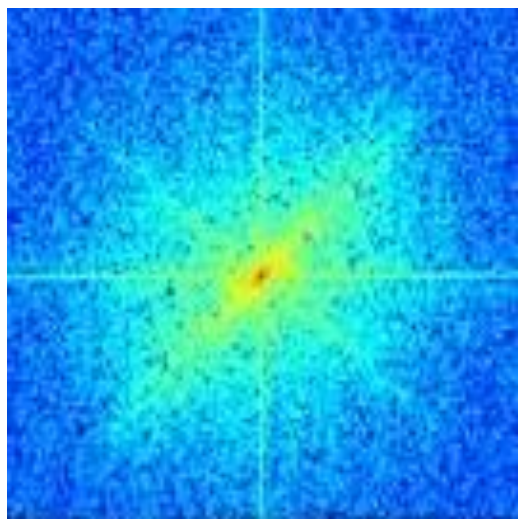
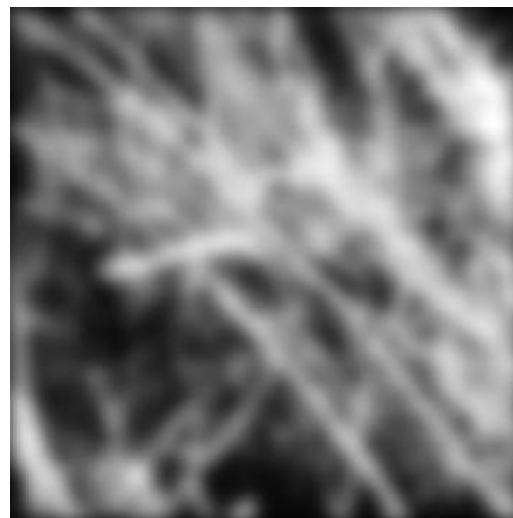
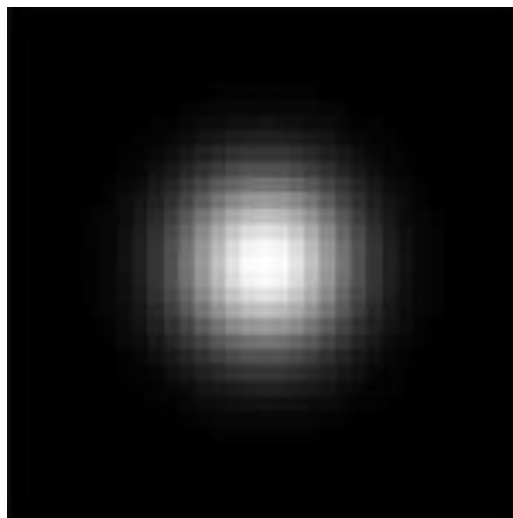
Gaussian
filter



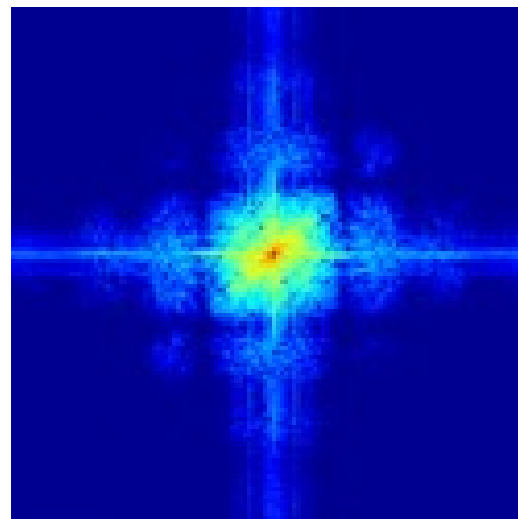
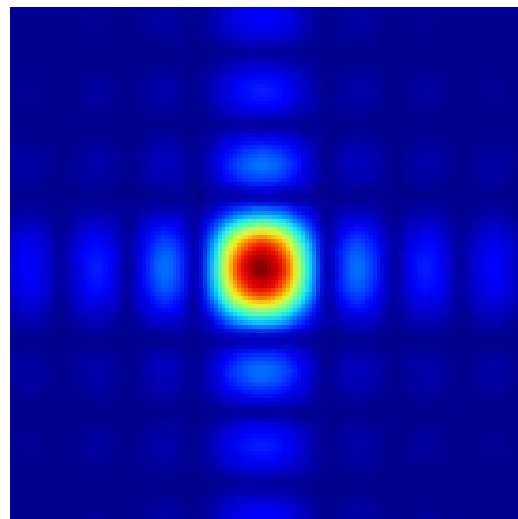
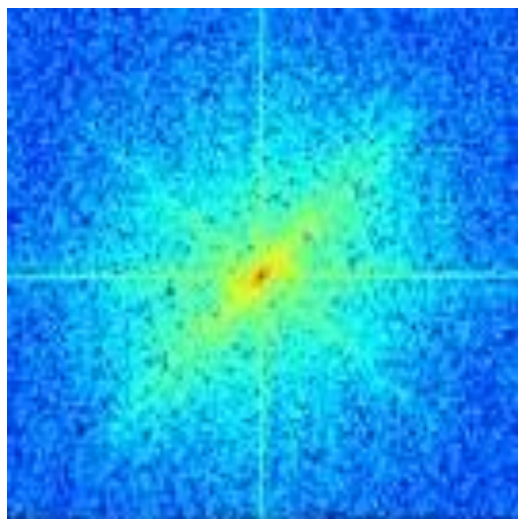
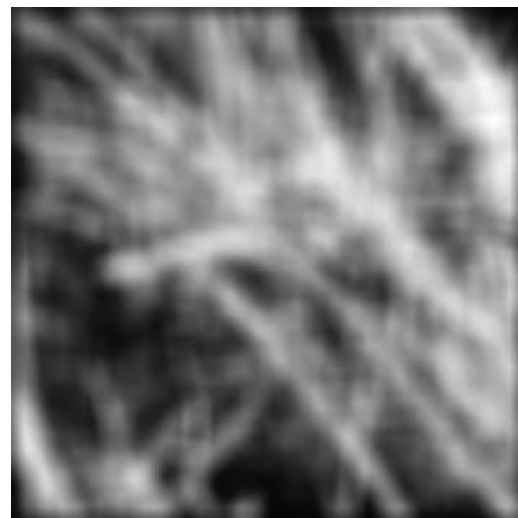
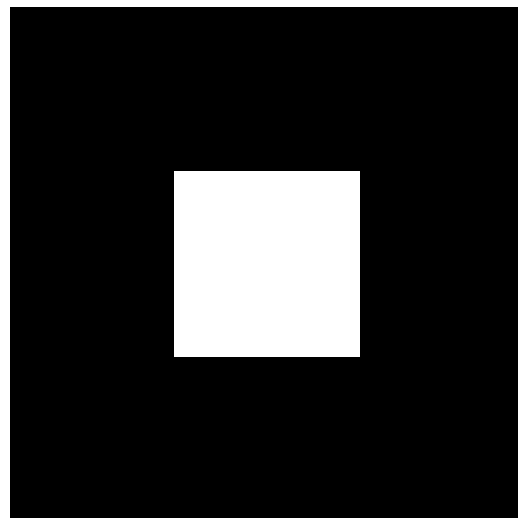
Box
filter



Gaussian blur



Box blur



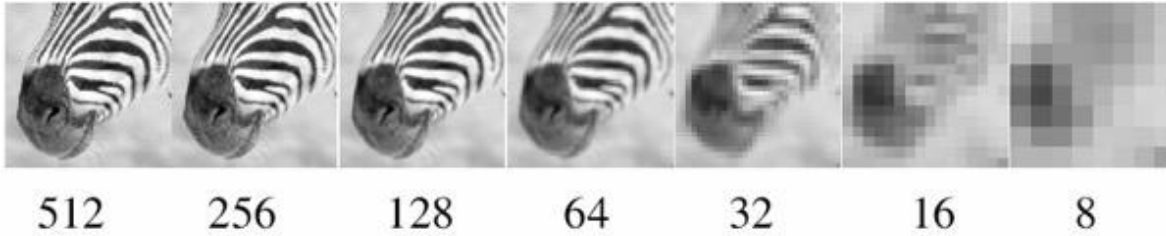
The Nyquist-Shannon sampling theorem

A continuous signal can be perfectly reconstructed from its discrete version using linear interpolation, if sampling occurred with frequency:

$$f_s \geq 2f_{\max} \quad \leftarrow \text{This is called the Nyquist frequency}$$

Equivalent reformulation: When downsampling, aliasing does not occur if samples are taken at the Nyquist frequency or higher.

Gaussian pyramid



How does the Nyquist-Shannon theorem relate to the Gaussian pyramid?



Finding corners

(a.k.a. PCA)

1. Compute image gradients over small region
2. Subtract mean from each image gradient
3. Compute the covariance matrix
4. Compute eigenvectors and eigenvalues
5. Use threshold on eigenvalues to detect corners

$$I_x = \frac{\partial I}{\partial x}$$



$$I_y = \frac{\partial I}{\partial y}$$



$$\begin{bmatrix} \sum_{p \in P} I_x I_x & \sum_{p \in P} I_x I_y \\ \sum_{p \in P} I_y I_x & \sum_{p \in P} I_y I_y \end{bmatrix}$$

Error function

Change of intensity for the shift $[u,v]$:

$$E(u, v) = \sum_{x, y} w(x, y) [I(x + u, y + v) - I(x, y)]^2$$

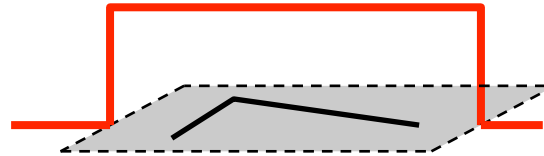
Error
function

Window
function

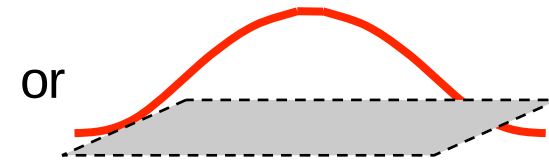
Shifted
intensity

Intensity

Window function $w(x, y) =$



1 in window, 0 outside



Gaussian

Bilinear approximation

For small shifts $[u, v]$ we have a 'bilinear approximation':

Change in
appearance for a
shift $[u, v]$

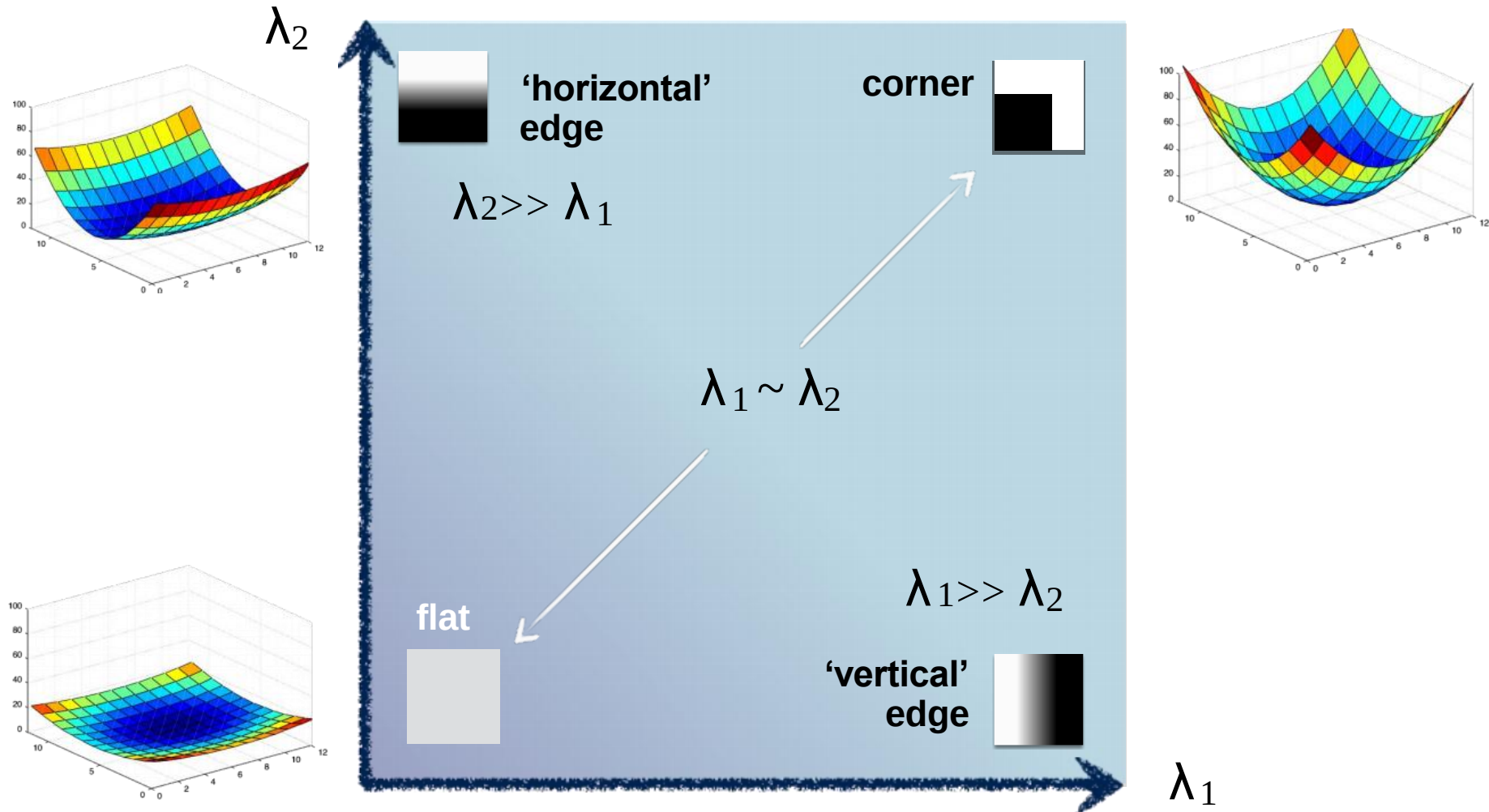
$$E(u, v) \cong [u, v] M \begin{bmatrix} u \\ v \end{bmatrix}$$

where M is a 2×2 matrix computed from image derivatives:

'second moment' matrix
'structure tensor'

$$M = \sum_{x, y} w(x, y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

interpreting eigenvalues



Harris & Stephens (1988)

$$R = \det(M) - \kappa \text{trace}^2(M)$$

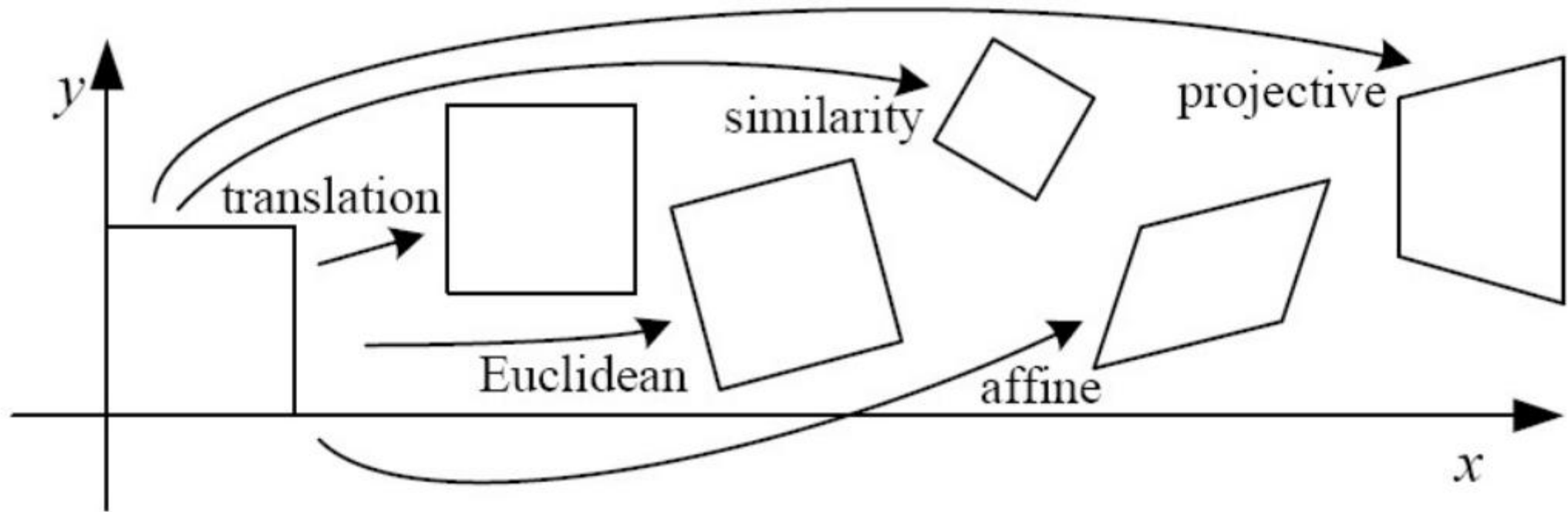
Kanade & Tomasi (1994)

$$R = \min(\lambda_1, \lambda_2)$$

Nobel (1998)

$$R = \frac{\det(M)}{\text{trace}(M) + \epsilon}$$

Classification of 2D transformations

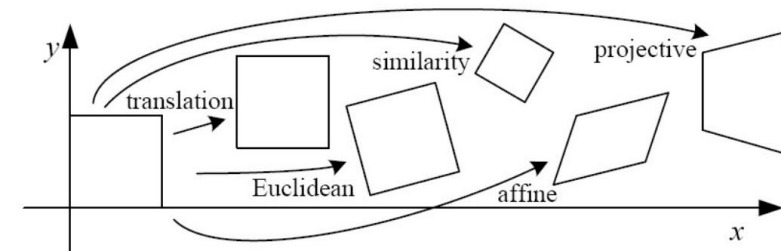


Classification of 2D transformations

Euclidean (rigid):
rotation + translation

$$\begin{bmatrix} \cos \theta & -\sin \theta & r_3 \\ \sin \theta & \cos \theta & r_6 \\ 0 & 0 & 1 \end{bmatrix}$$

How many degrees of freedom?



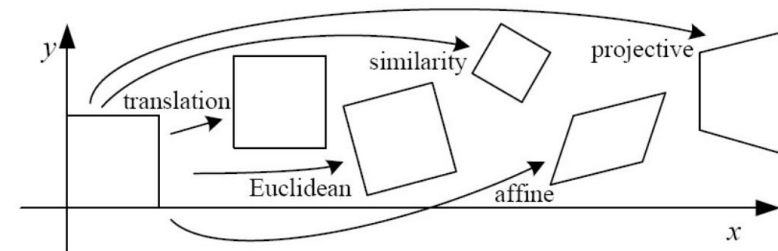
Classification of 2D transformations

Affine transform:
uniform scaling + shearing
+ rotation + translation

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ 0 & 0 & 1 \end{bmatrix}$$

How many degrees of freedom?

$$\begin{array}{cc} \text{similarity} & \text{shear} \\ \begin{bmatrix} sr_1 & sr_2 \\ sr_3 & sr_4 \end{bmatrix} & \begin{bmatrix} 1 & h_1 \\ h_2 & 1 \end{bmatrix} \end{array} = \begin{bmatrix} sr_1 + h_2 sr_2 & sr_2 + h_1 sr_1 \\ sr_3 + h_2 sr_4 & sr_4 + h_1 sr_3 \end{bmatrix}$$



Projective transformations (aka homographies)

Projective transformations are combinations of

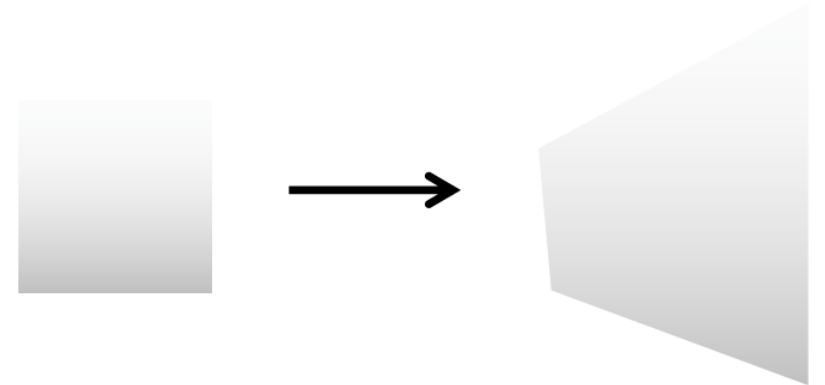
- affine transformations; and
- projective wraps

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

Properties of projective transformations:

- origin does not necessarily map to origin
- lines map to lines
- parallel lines do not necessarily map to parallel lines
- ratios are not necessarily preserved
- compositions of projective transforms are also projective transforms

8 DOF: vectors (and therefore matrices) are defined up to scale)



Determining the homography matrix

Write out linear equation for each correspondence:

$$P' = H \cdot P \quad \text{or} \quad \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} h_1 & h_2 & h_3 \\ h_4 & h_5 & h_6 \\ h_7 & h_8 & h_9 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Expand matrix multiplication:

$$x' = \alpha(h_1x + h_2y + h_3)$$

$$y' = \alpha(h_4x + h_5y + h_6)$$

$$1 = \alpha(h_7x + h_8y + h_9)$$

Divide out unknown scale factor:

$$x'(h_7x + h_8y + h_9) = (h_1x + h_2y + h_3)$$

$$y'(h_7x + h_8y + h_9) = (h_4x + h_5y + h_6)$$

*How do you
rearrange terms
to make it a
linear system?*

$$x'(h_7x + h_8y + h_9) = (h_1x + h_2y + h_3)$$

$$y'(h_7x + h_8y + h_9) = (h_4x + h_5y + h_6)$$

Just rearrange the terms



$$h_7xx' + h_8yx' + h_9x' - h_1x - h_2y - h_3 = 0$$

$$h_7xy' + h_8yy' + h_9y' - h_4x - h_5y - h_6 = 0$$

Determining the homography matrix

Stack together constraints from multiple point correspondences:

$$\mathbf{A}\mathbf{h} = \mathbf{0}$$

$$\begin{bmatrix} -x & -y & -1 & 0 & 0 & 0 & xx' & yx' & x' \\ 0 & 0 & 0 & -x & -y & -1 & xy' & yy' & y' \\ -x & -y & -1 & 0 & 0 & 0 & xx' & yx' & x' \\ 0 & 0 & 0 & -x & -y & -1 & xy' & yy' & y' \\ \vdots & & & & & & & & \\ -x & -y & -1 & 0 & 0 & 0 & xx' & yx' & x' \\ 0 & 0 & 0 & -x & -y & -1 & xy' & yy' & y' \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \\ h_6 \\ h_7 \\ h_8 \\ h_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Homogeneous linear least squares problem

General form of total least squares

(**Warning:** change of notation. $\mathbf{x}$ is a vector of parameters!)

$$\begin{aligned} E_{\text{TLS}} &= \sum_i (\mathbf{a}_i \mathbf{x})^2 \\ &= \|\mathbf{A}\mathbf{x}\|^2 && \text{(matrix form)} \\ \|\mathbf{x}\|^2 &= 1 && \text{constraint} \end{aligned}$$

minimize	$\ \mathbf{A}\mathbf{x}\ ^2$		(Rayleigh quotient)
subject to	$\ \mathbf{x}\ ^2 = 1$		minimize $\frac{\ \mathbf{A}\mathbf{x}\ ^2}{\ \mathbf{x}\ ^2}$

Solution is the eigenvector
corresponding to smallest
eigenvalue of

$$\mathbf{A}^\top \mathbf{A}$$

(equivalent)

Solution is the column of $\mathbf{V}$
corresponding to smallest singular
value

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

Singular value decomposition

$$\underset{n \times m}{\mathbf{A}} = \overset{\substack{\text{orthonormal} \\ \downarrow}}{\underset{n \times n}{\mathbf{U}}} \overset{\substack{\text{diagonal} \\ \downarrow}}{\underset{n \times m}{\mathbf{\Sigma}}} \overset{\substack{\text{orthonormal} \\ \swarrow}}{\underset{m \times m}{\mathbf{V}}}^T$$

$$= \sum_{i=1}^9 \sigma_i \underset{n \times 1}{\mathbf{u}_i} \underset{1 \times m}{\mathbf{v}_i}^T$$

Solving for H using DLT

Given $\{x_i, x'_i\}$ solve for H such that $x' = Hx$

1. For each correspondence, create 2x9 matrix A_i
2. Concatenate into single $2n \times 9$ matrix A
3. Compute SVD of $A = U\Sigma V^T$
4. Store singular vector of the smallest singular value $h = v_{\hat{i}}$
5. Reshape to get H

Estimating homography using RANSAC

- RANSAC loop

1. Get four point correspondences (randomly)
 2. Compute H using DLT
 3. Count inliers
 4. Keep H if largest number of inliers
- 

- Recompute H using all inliers

祝大家期末顺利，考出好成绩！